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Kim et al.

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(54) **METHOD BY WHICH TERMINAL CHANGES RECEPTION TIMING IN WIRELESS COMMUNICATION SYSTEM SUPPORTING SIDE LINKS, AND APPARATUS THEREFOR**

(71) Applicant: **LG ELECTRONICS INC.**, Seoul (KR)

(72) Inventors: **Yeongjun Kim**, Seoul (KR); **Youngdae Kim**, Seoul (KR); **Jungyong Park**, Seoul (KR)

(73) Assignee: **LG ELECTRONICS INC.**, Seoul (KR)

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H04W 72/0453 (2023.01)

H04W 76/14 (2018.01)

(52) **U.S. Cl.**

CPC **H04W 56/005** (2013.01); **H04W 72/0453** (2013.01); **H04W 76/14** (2018.02)

(58) **Field of Classification Search**

None

See application file for complete search history.

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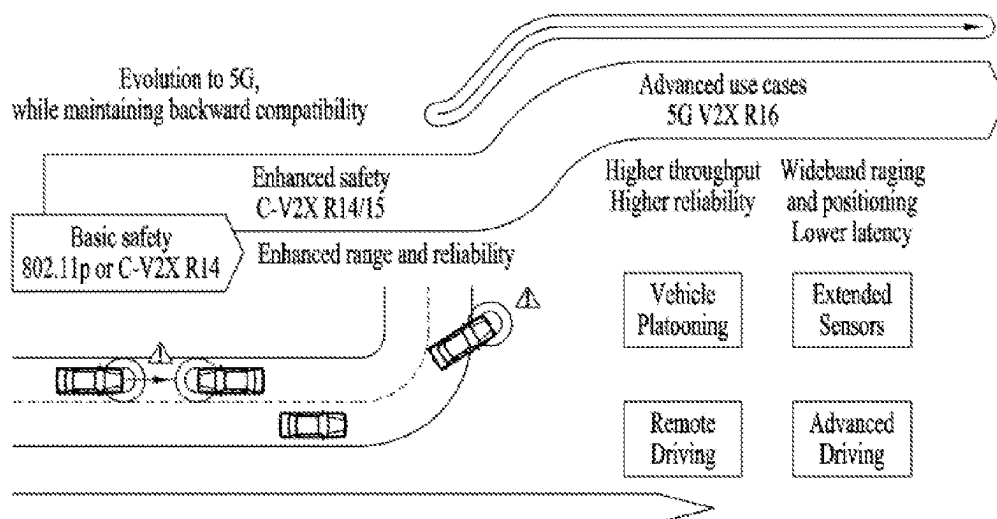
Primary Examiner — Jamaal Henson

(74) Attorney, Agent, or Firm — LEE, HONG, DEGERMAN, KANG & WAIMEY

(57) **ABSTRACT**

Disclosed are a method by which a first user equipment (UE) changes a reception timing in a wireless communication system supporting side links, and an apparatus therefor, according to various embodiments. The method comprises the steps of: forming a first side link with a second UE in a first frequency band; estimating a second reception timing related to a second side link formed in a second frequency band, which is adjacent to the first frequency band, on the basis of a synchronization signal for a third UE using the second frequency band, and timing advance (TA) information; and determining whether to change a reception timing of a side link signal for the first side link on the basis of a difference value between a first reception timing related to the first side link and the second reception timing.

14 Claims, 24 Drawing Sheets



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FIG. 1

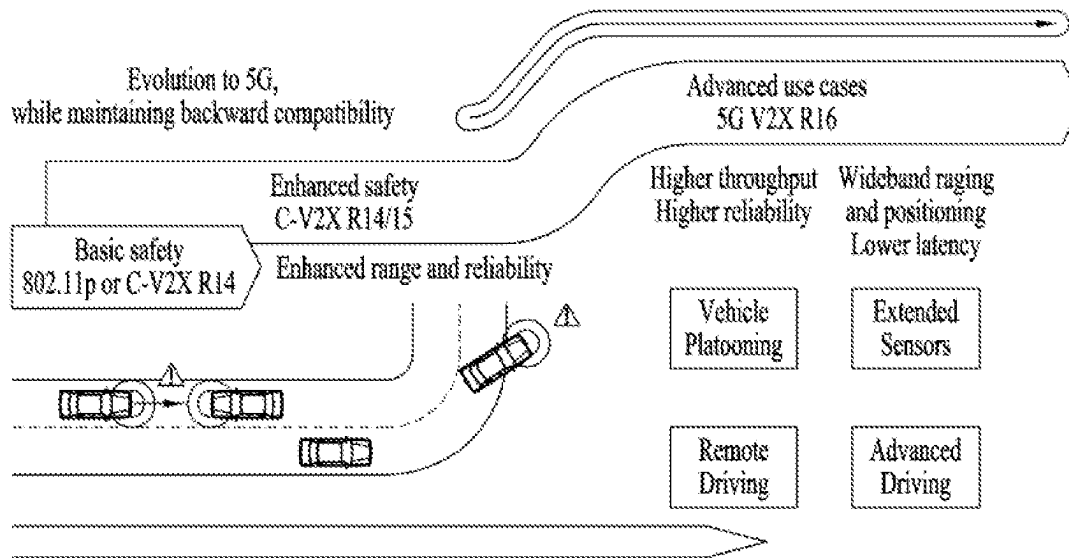


FIG. 2

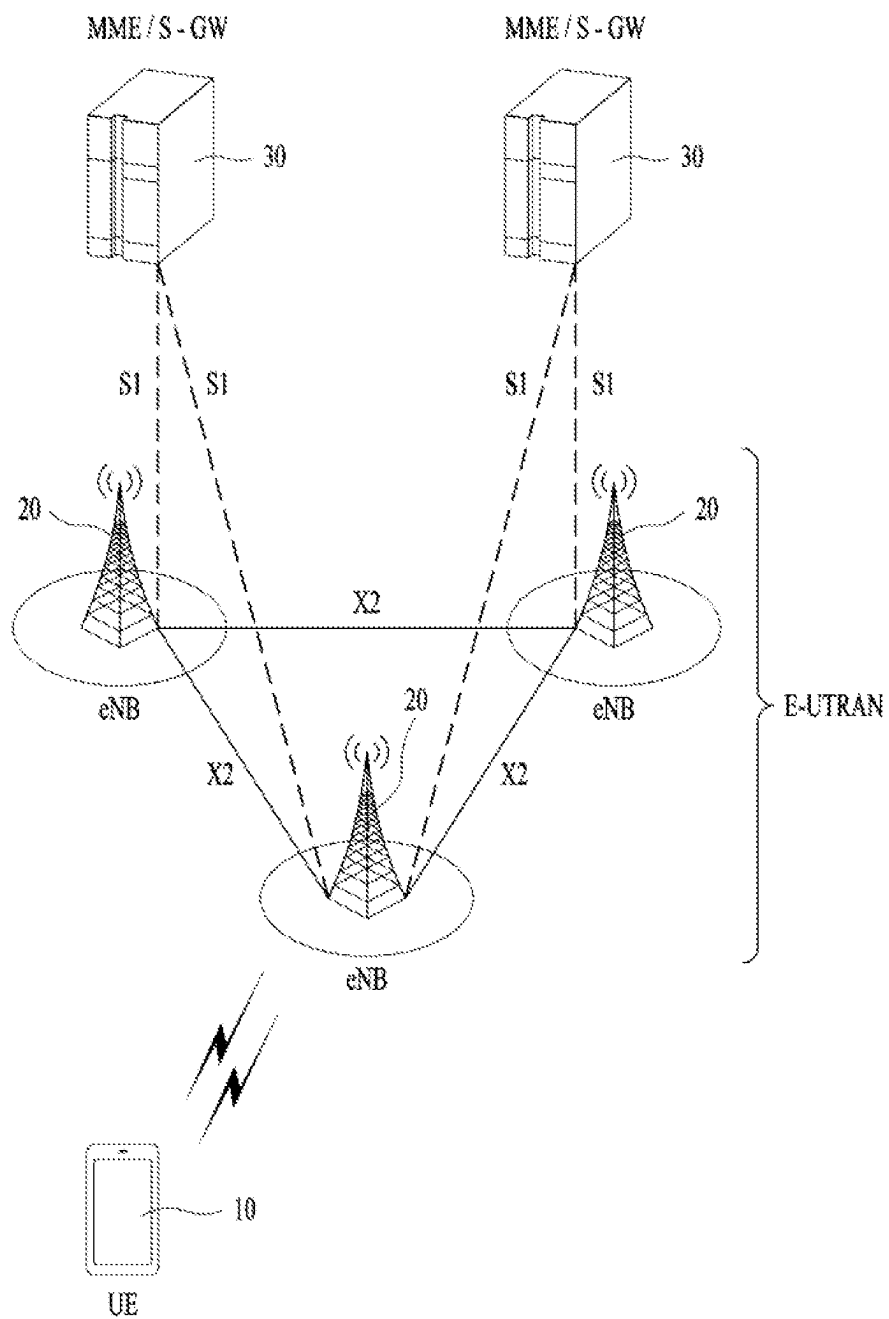


FIG. 3

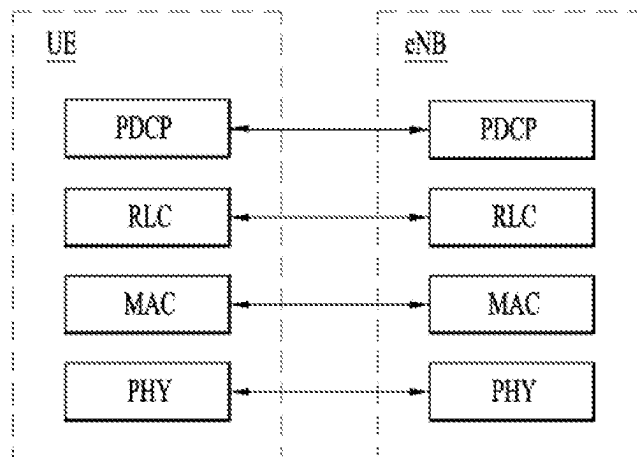


FIG. 4

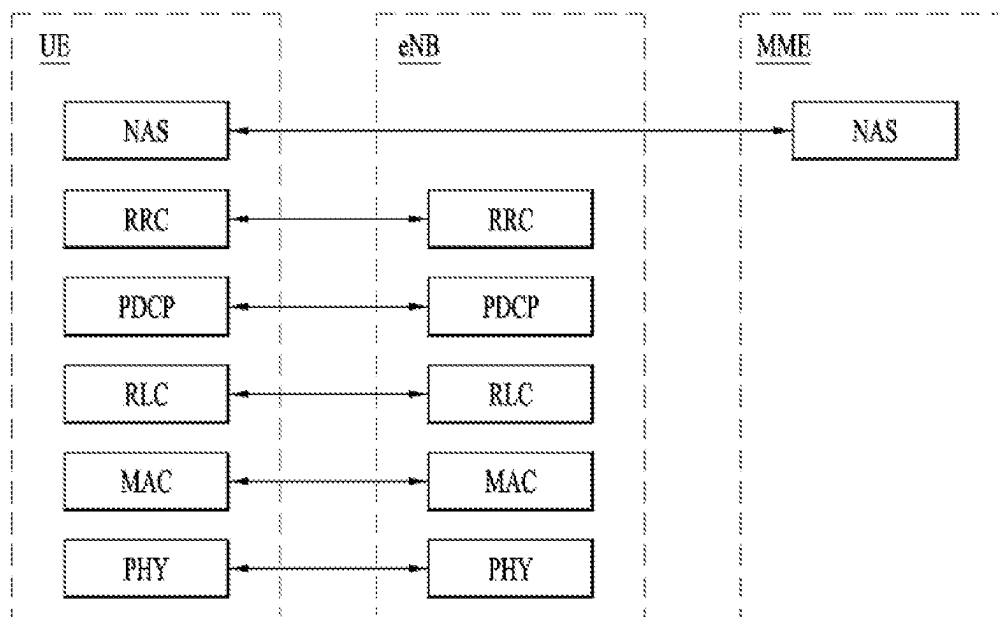


FIG. 5

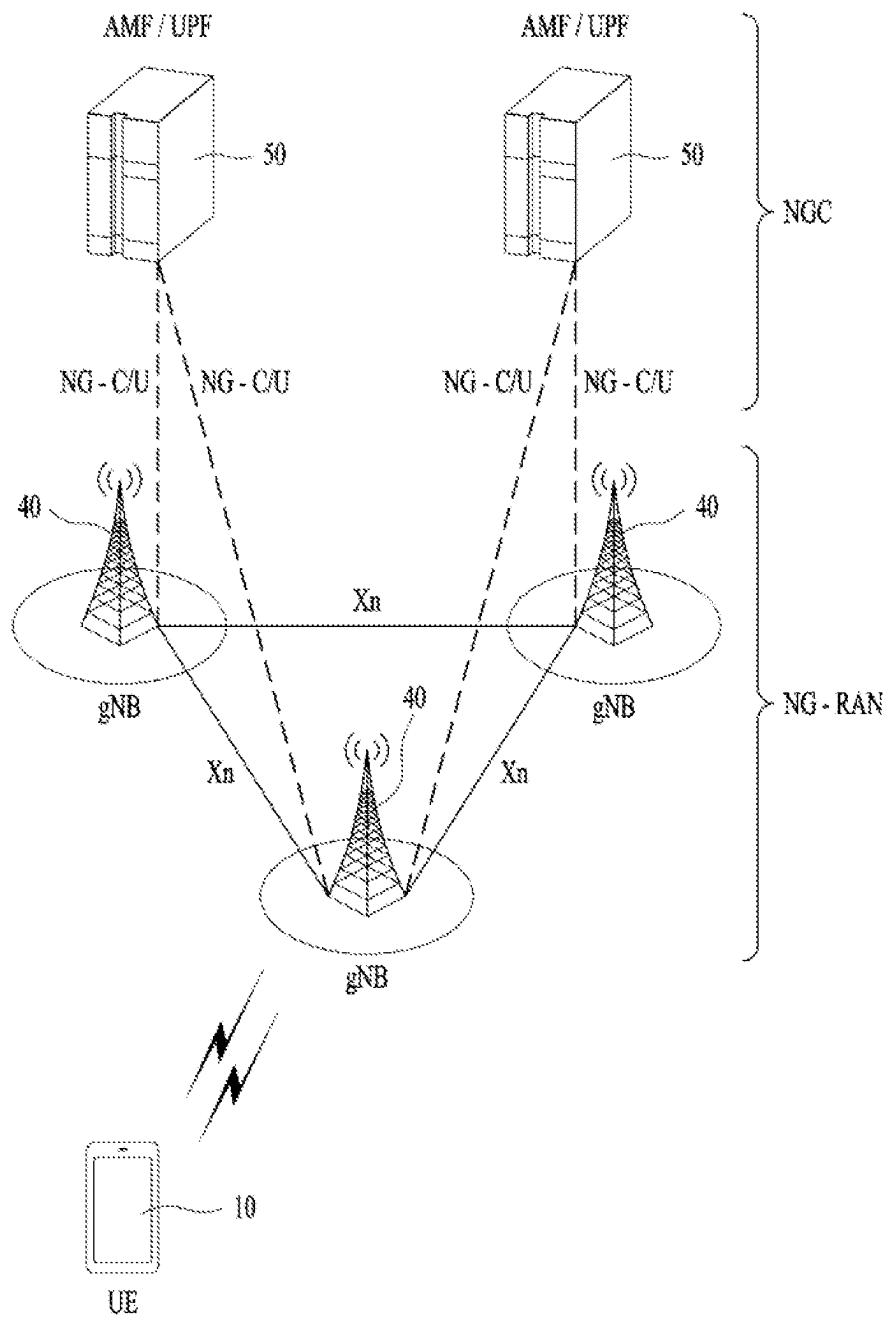


FIG. 6

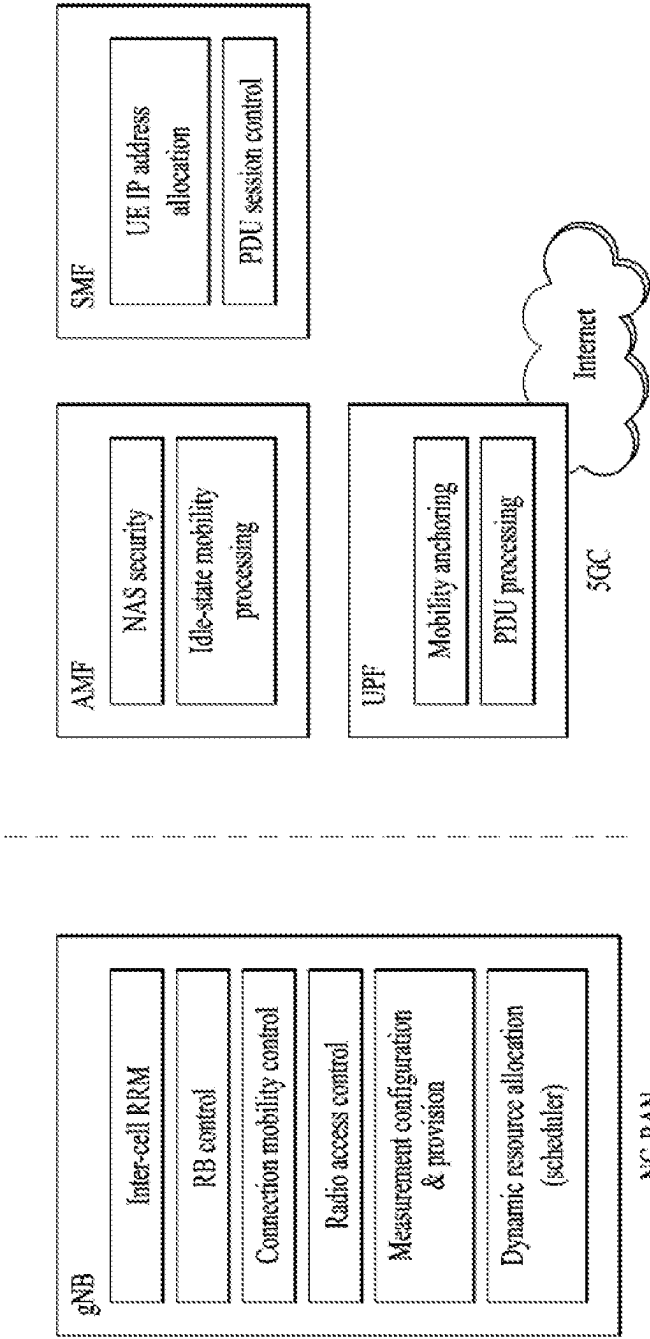


FIG. 7

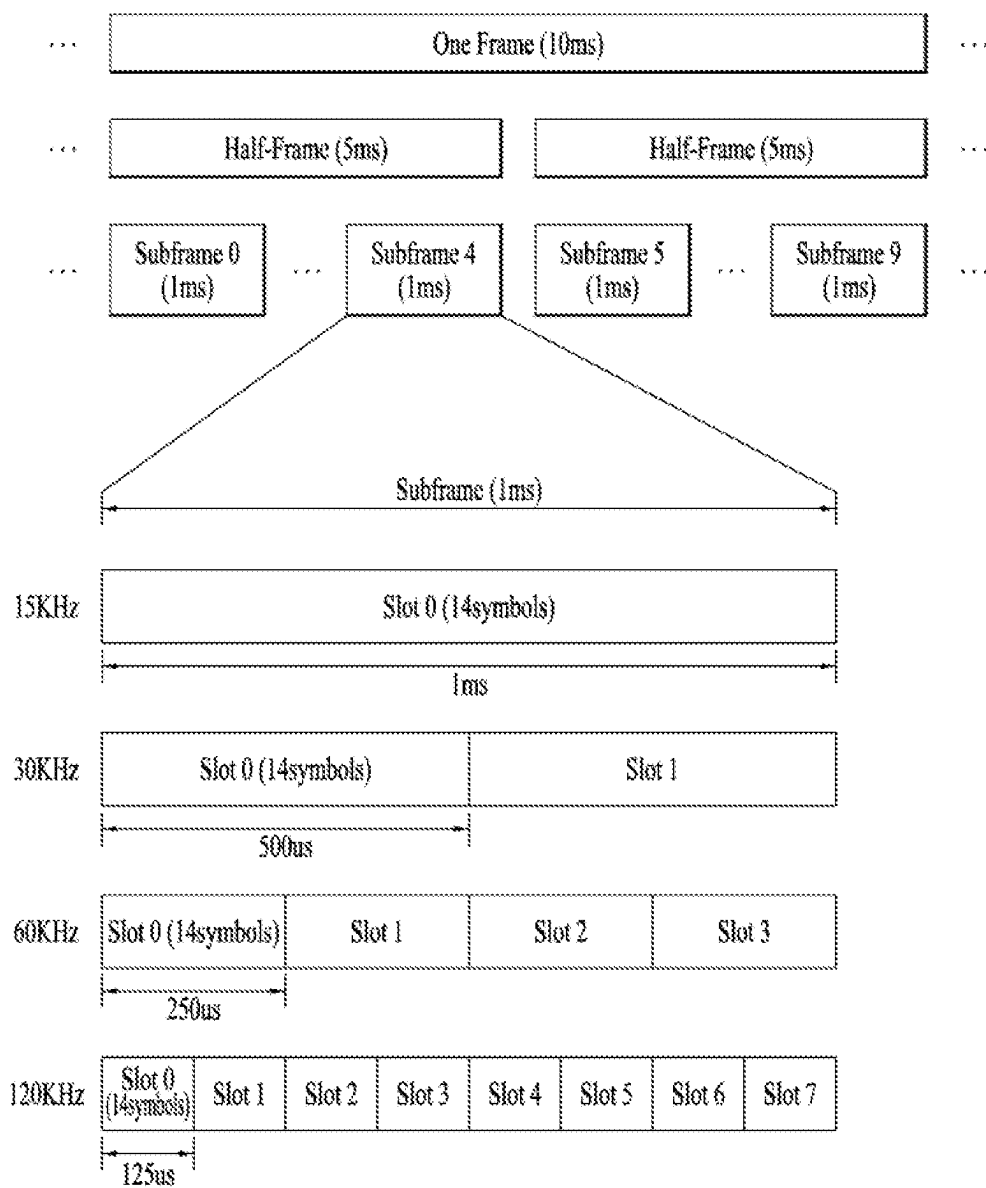


FIG. 8

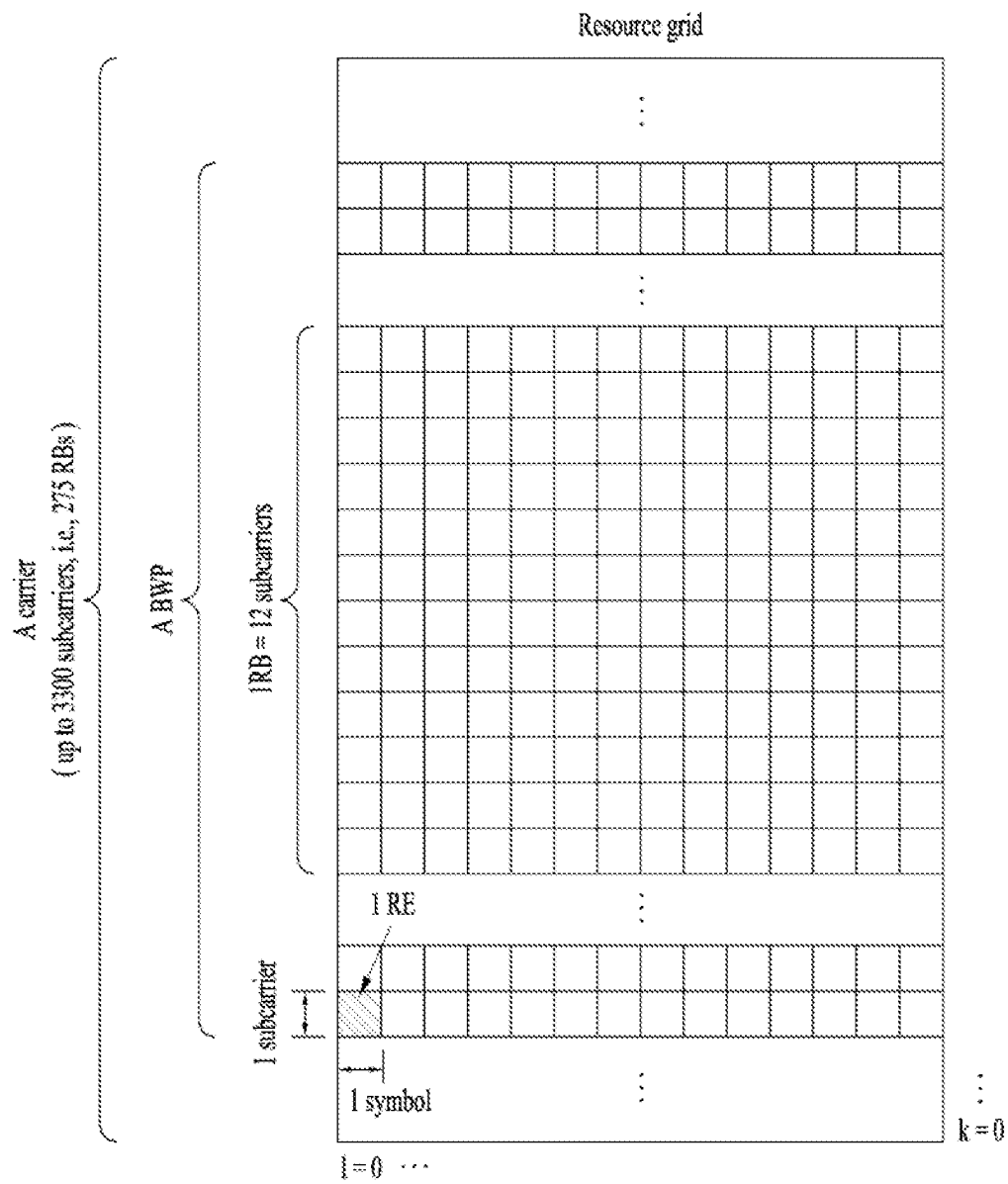


FIG. 9

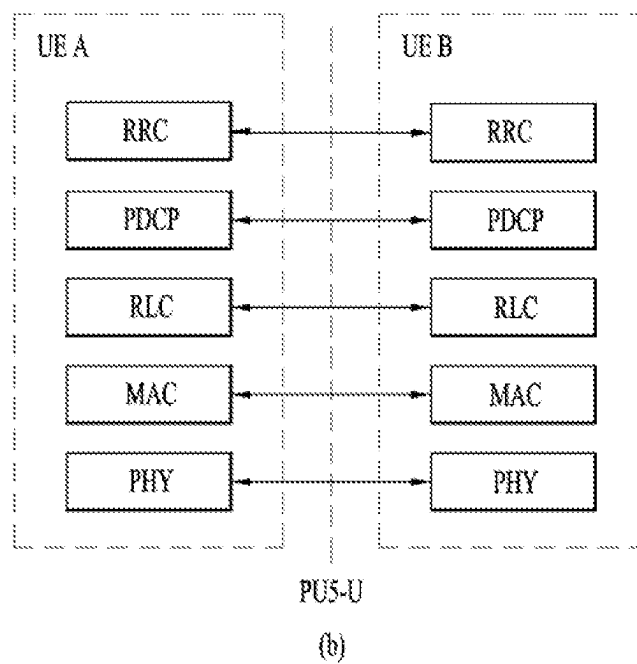
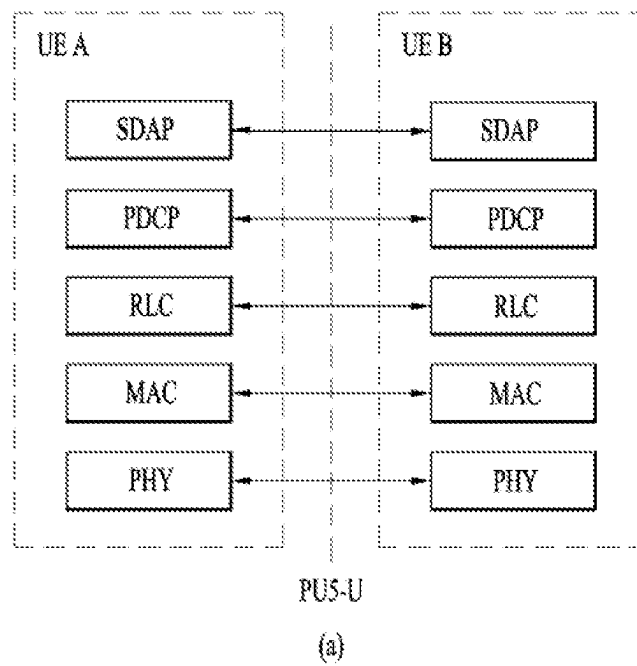


FIG. 11

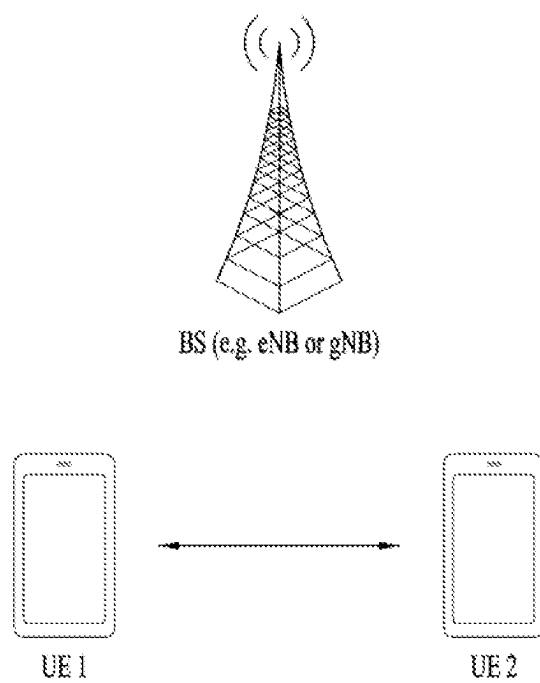


FIG. 12

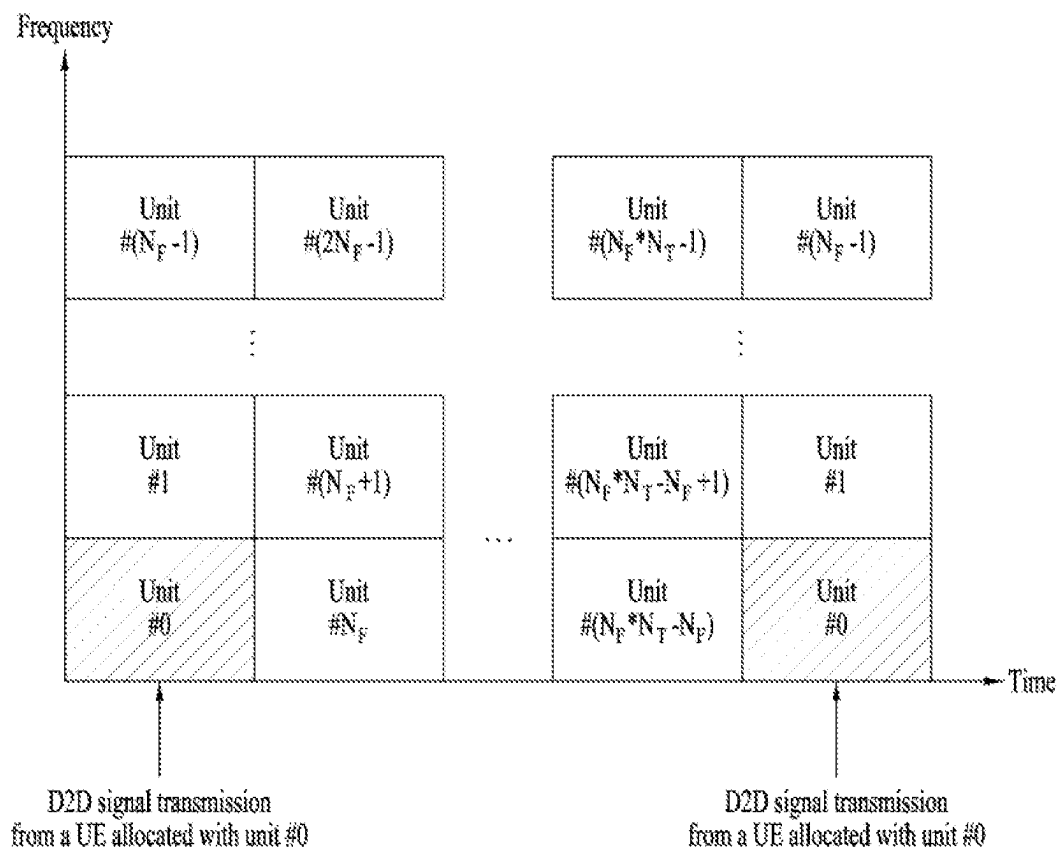
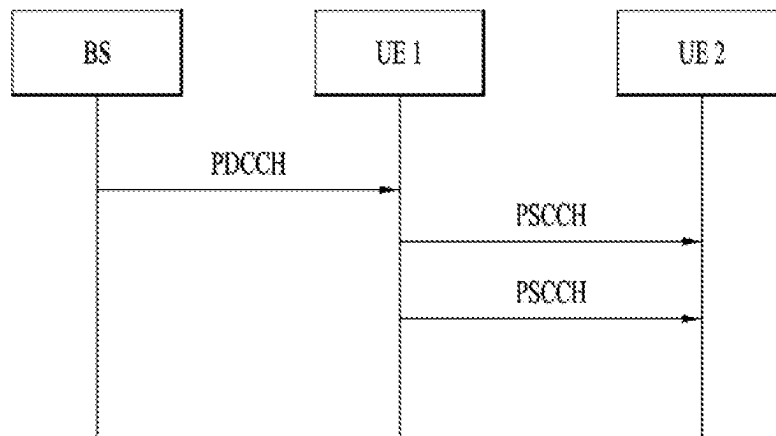
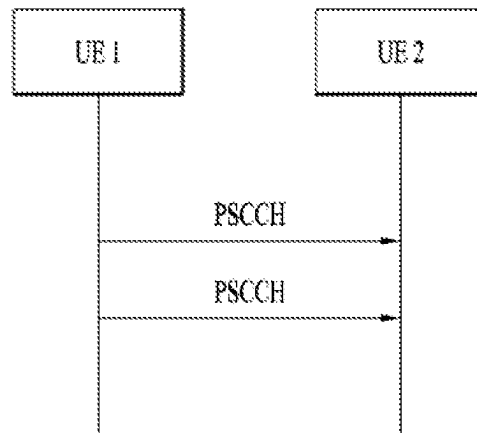


FIG. 13



(a)



(b)

FIG. 14

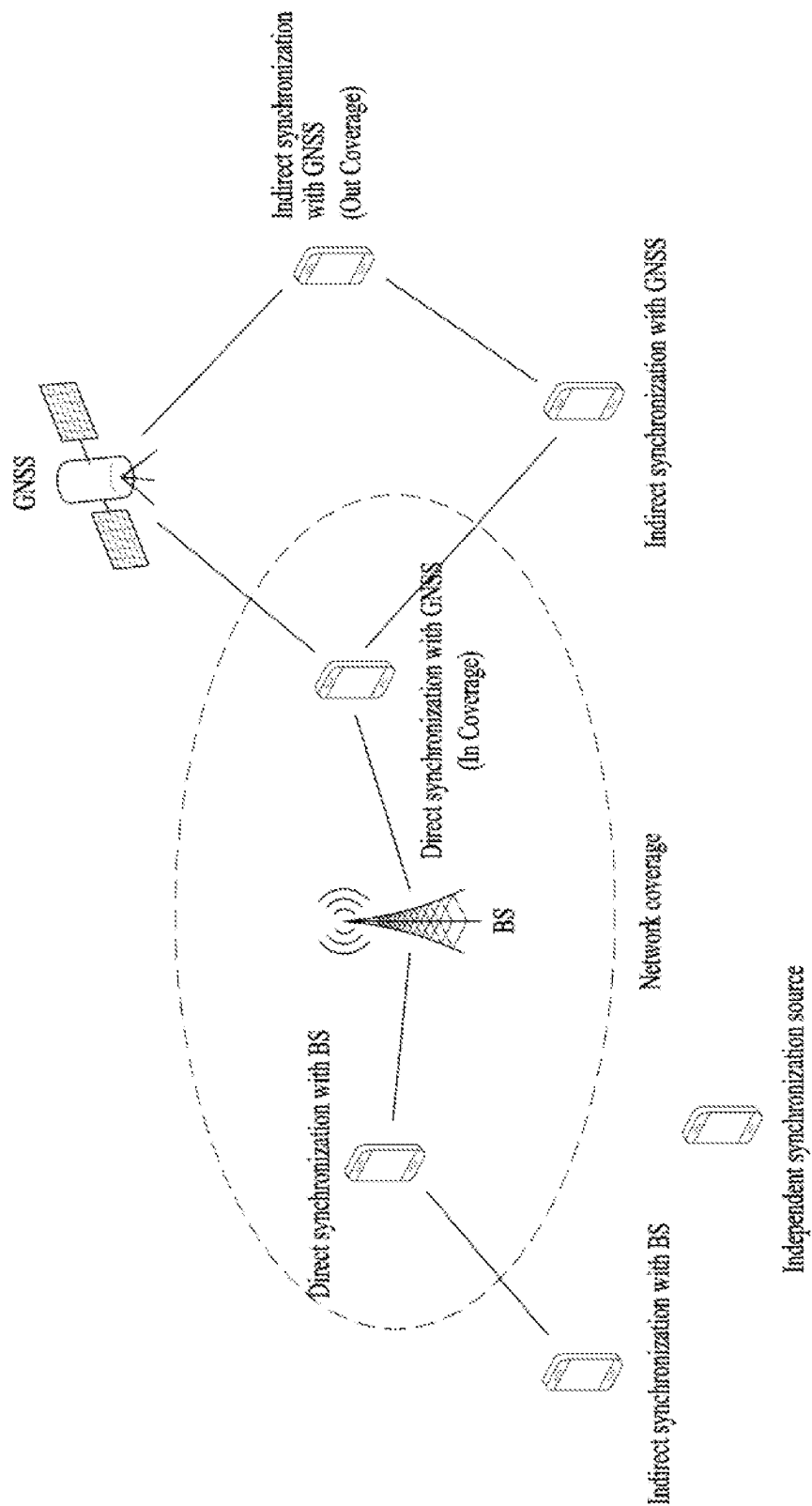


FIG. 15

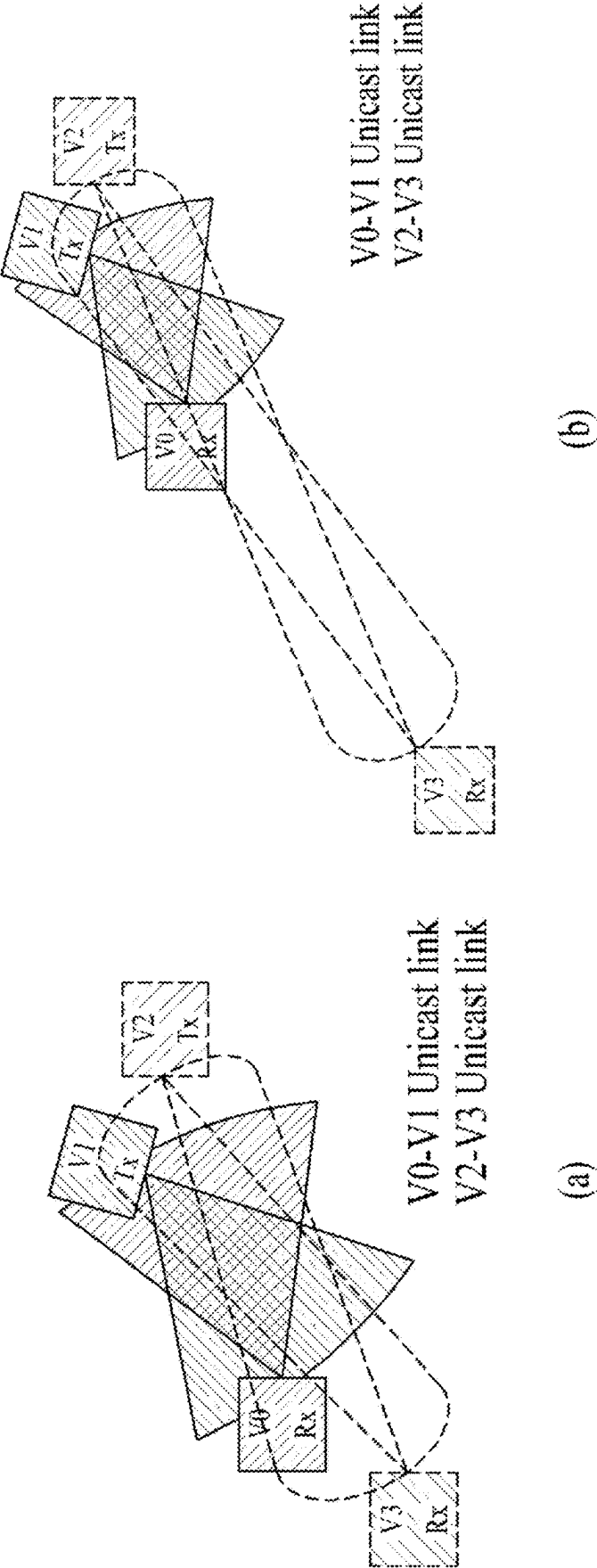


FIG. 16

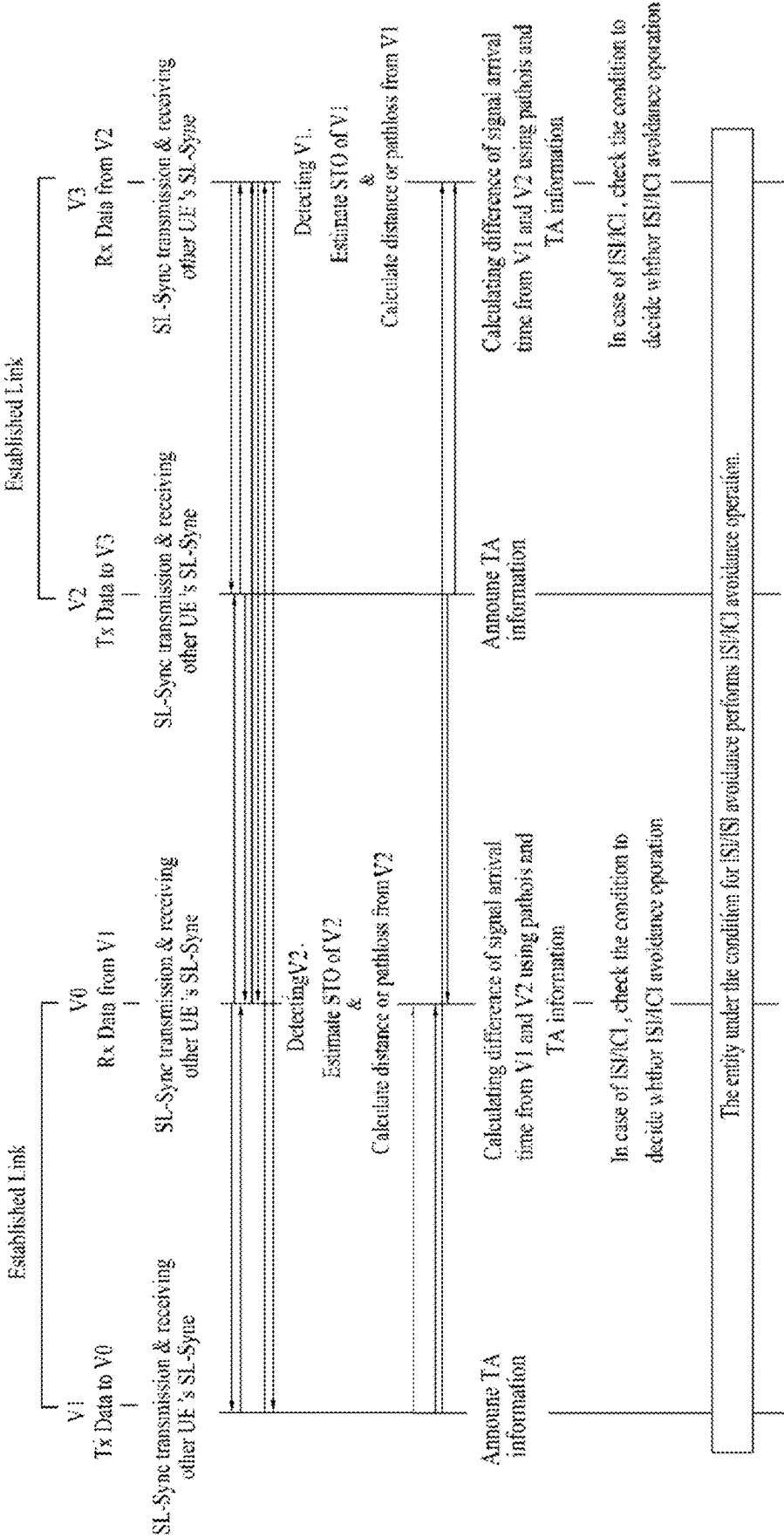


FIG. 17

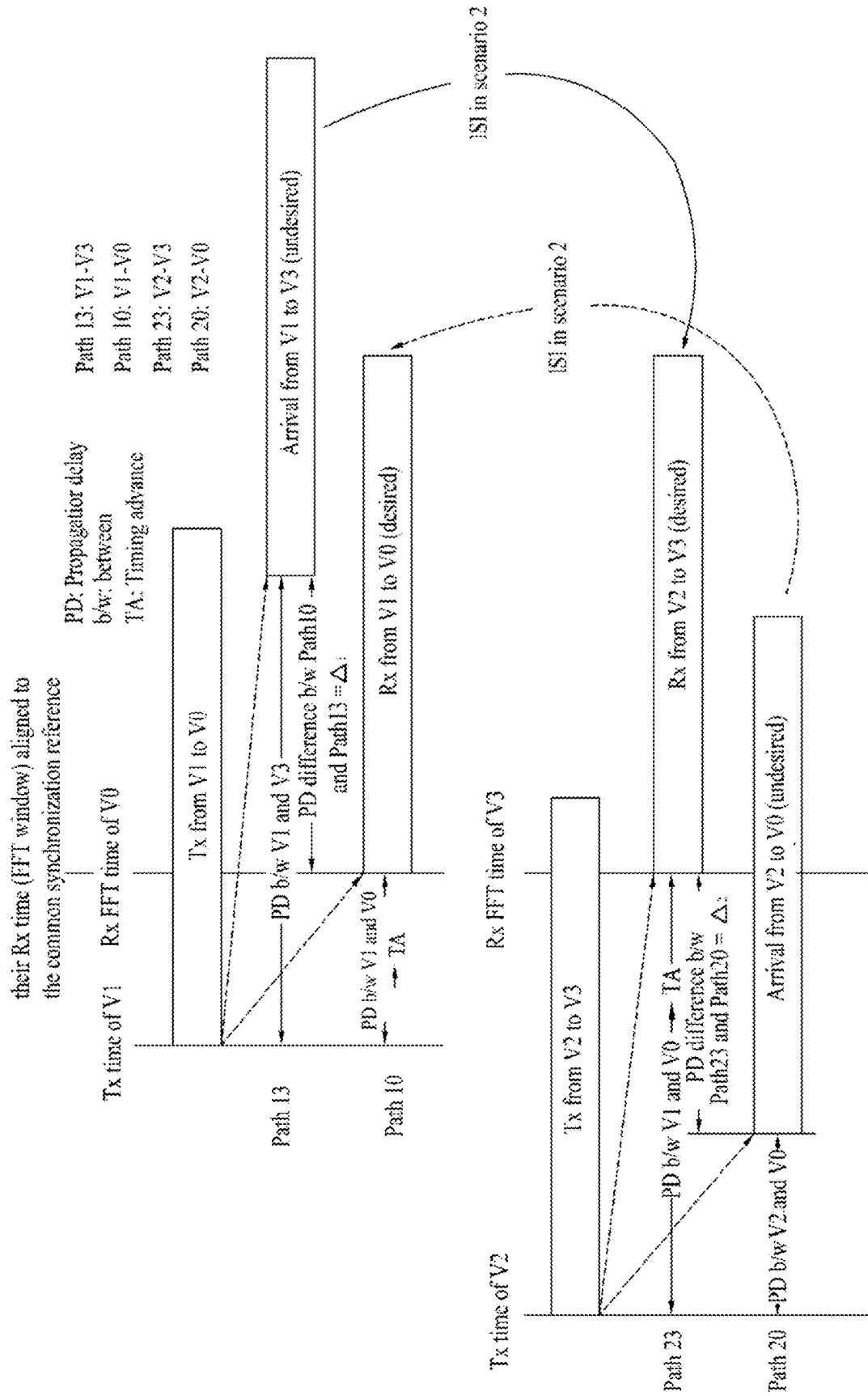


FIG. 18

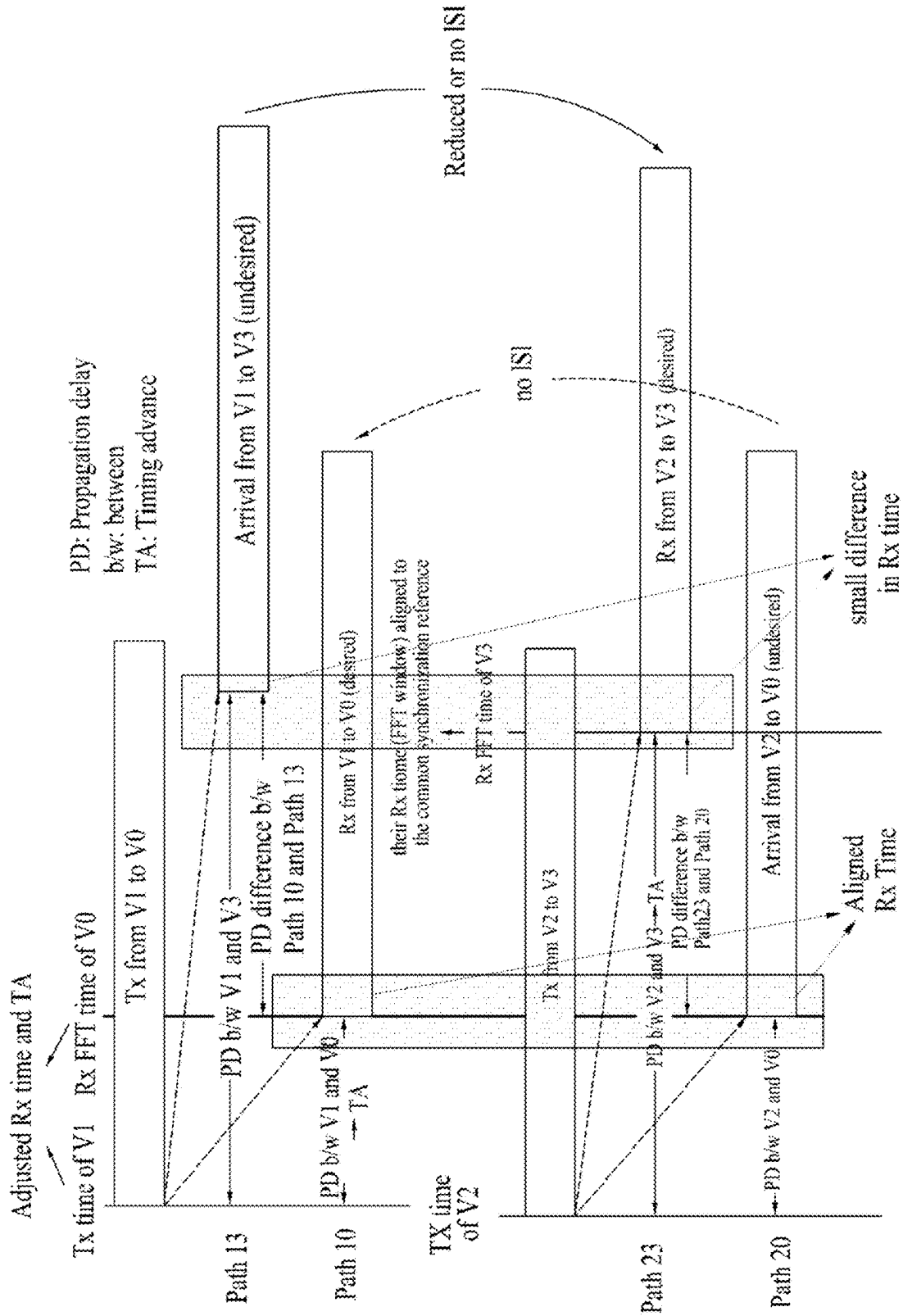


FIG. 19

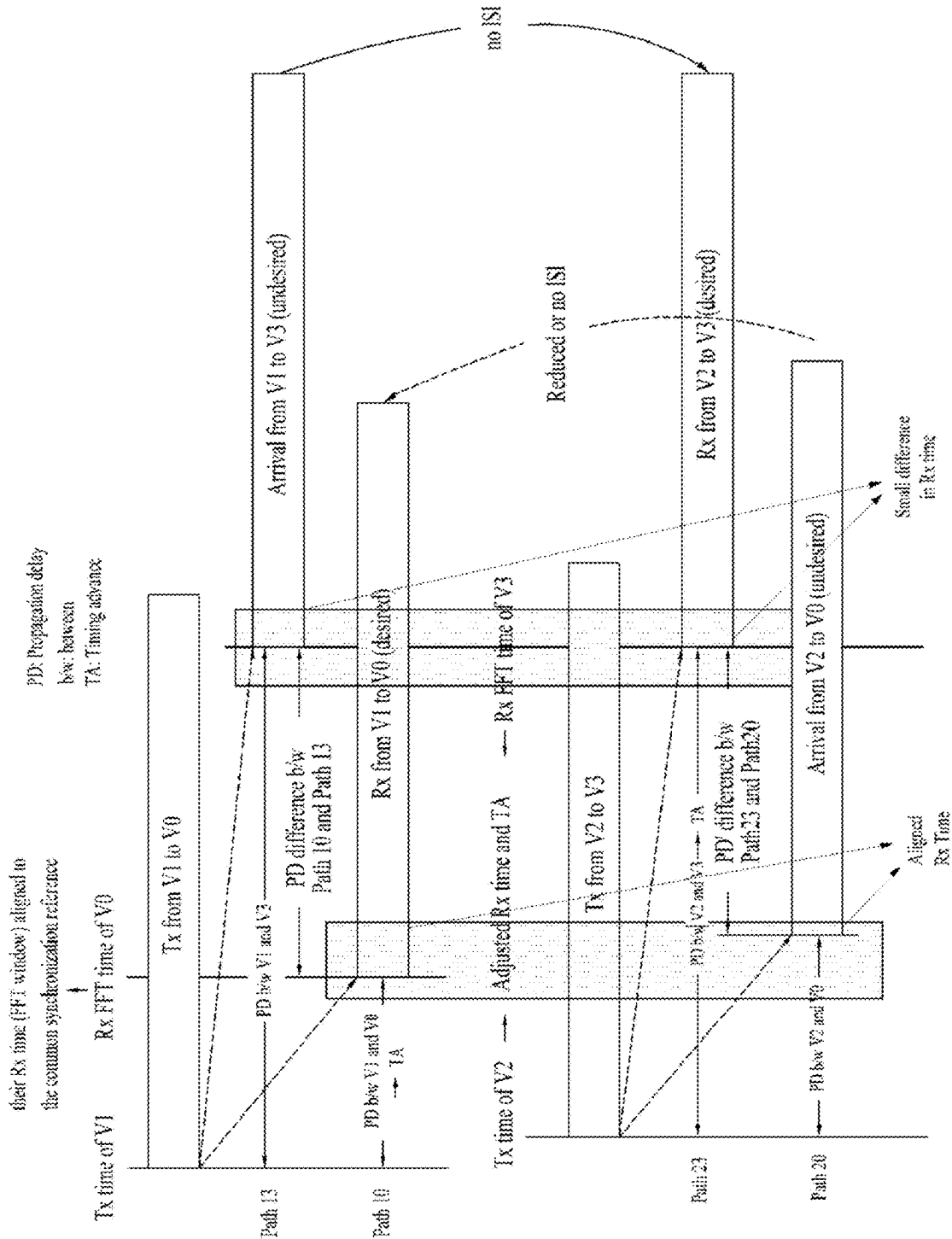


FIG. 20

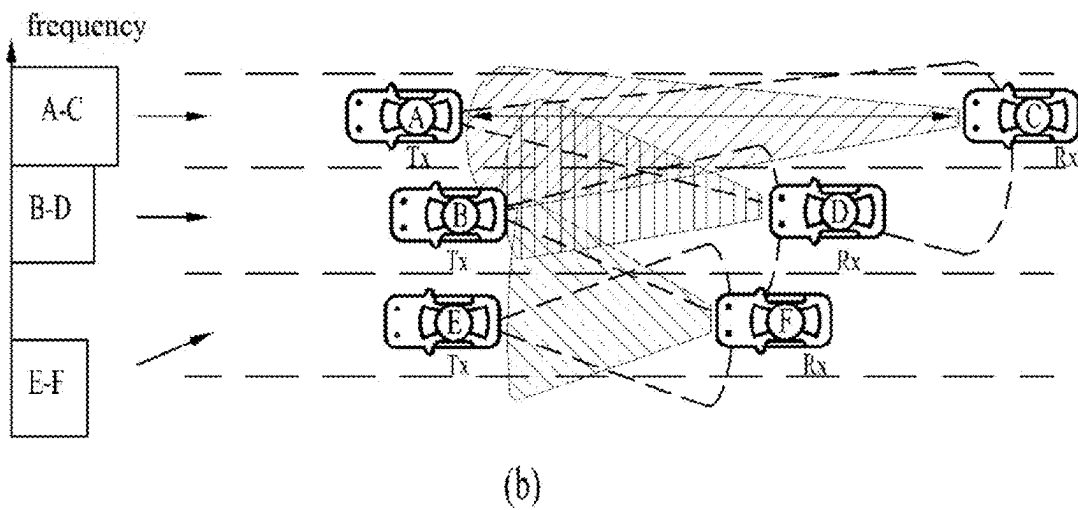
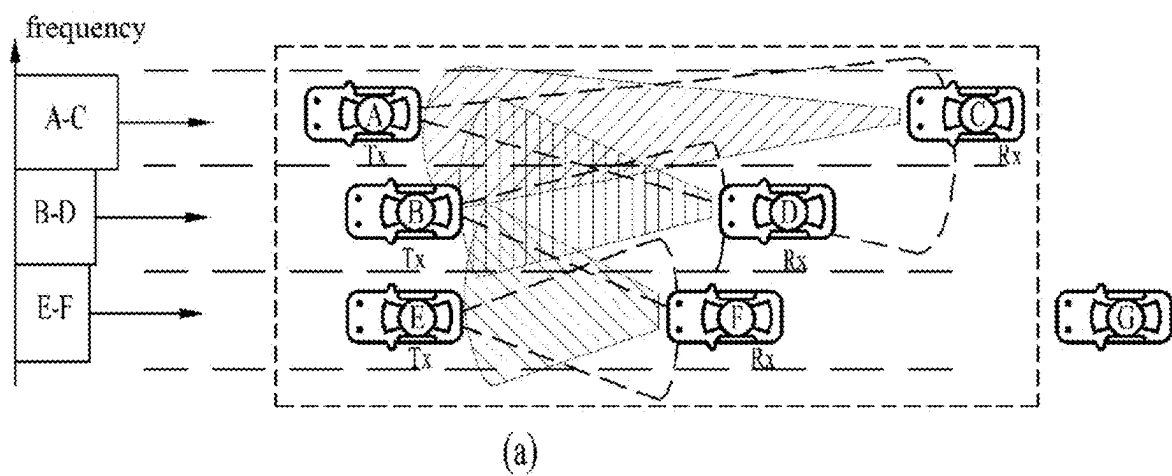


FIG. 21

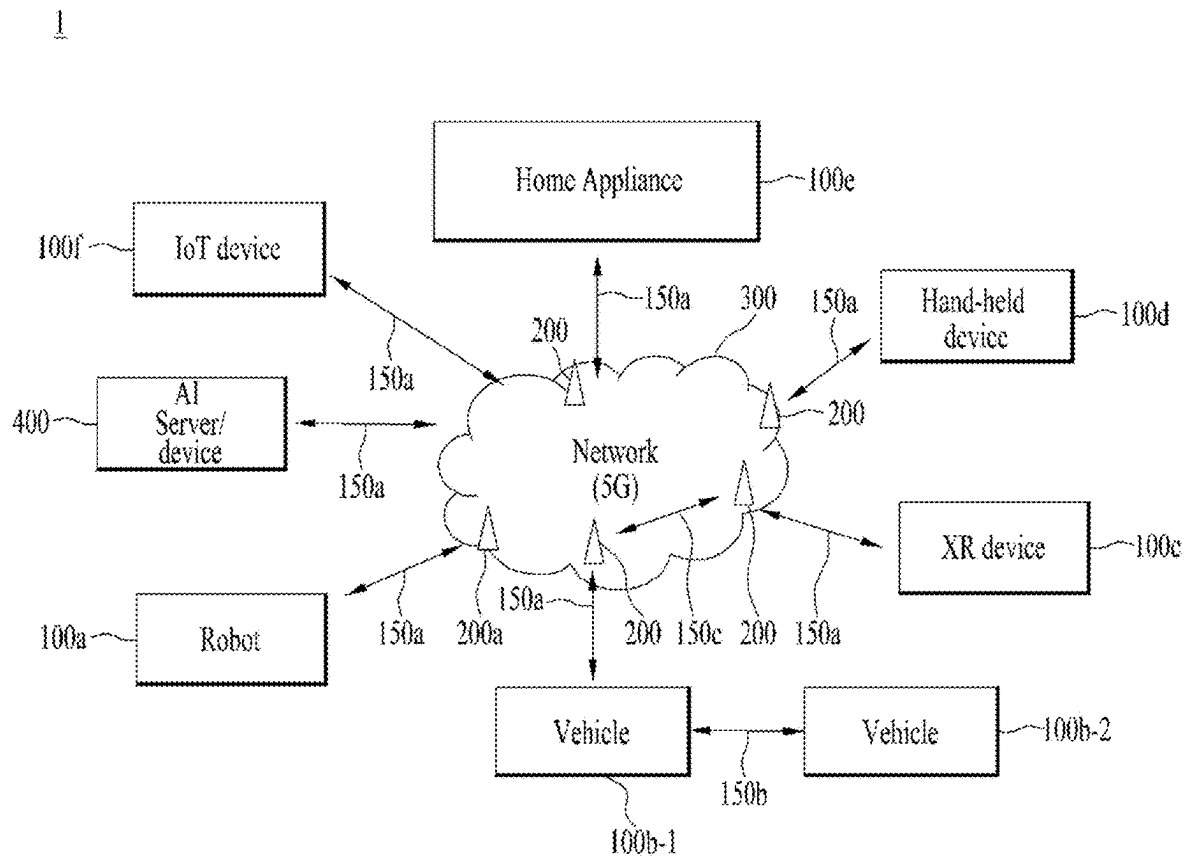


FIG. 22

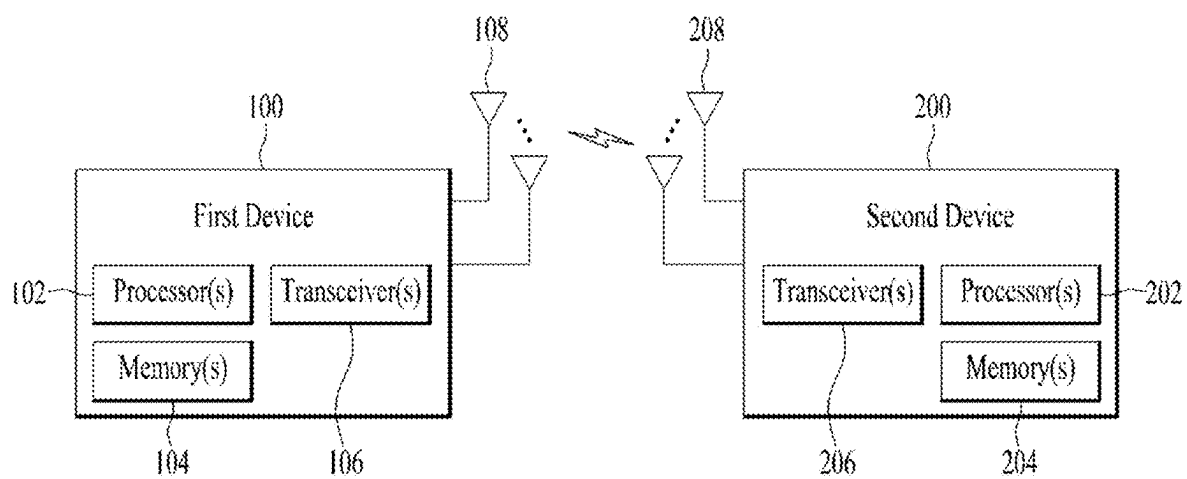


FIG. 23

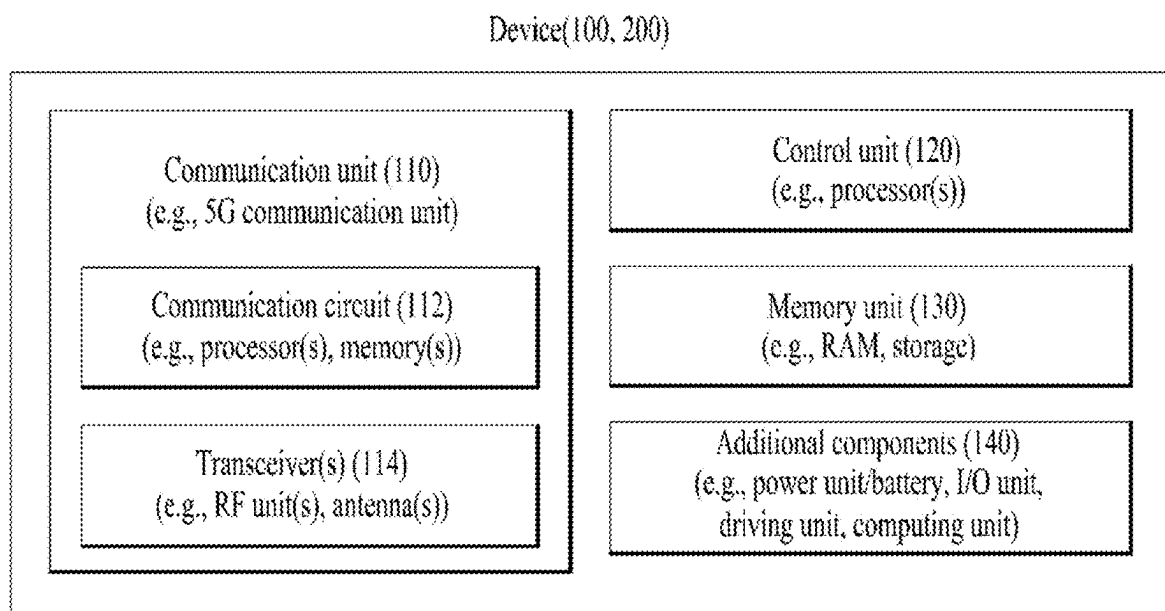


FIG. 24

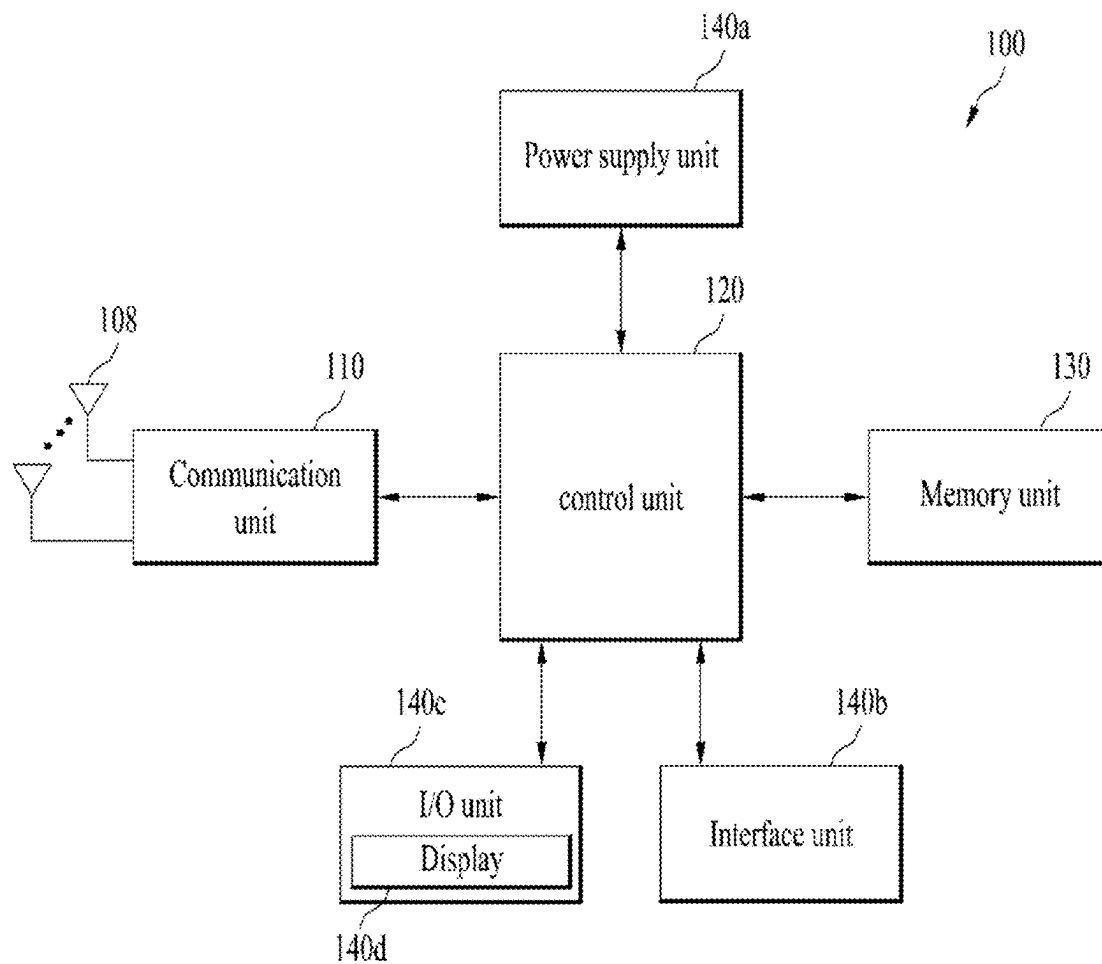
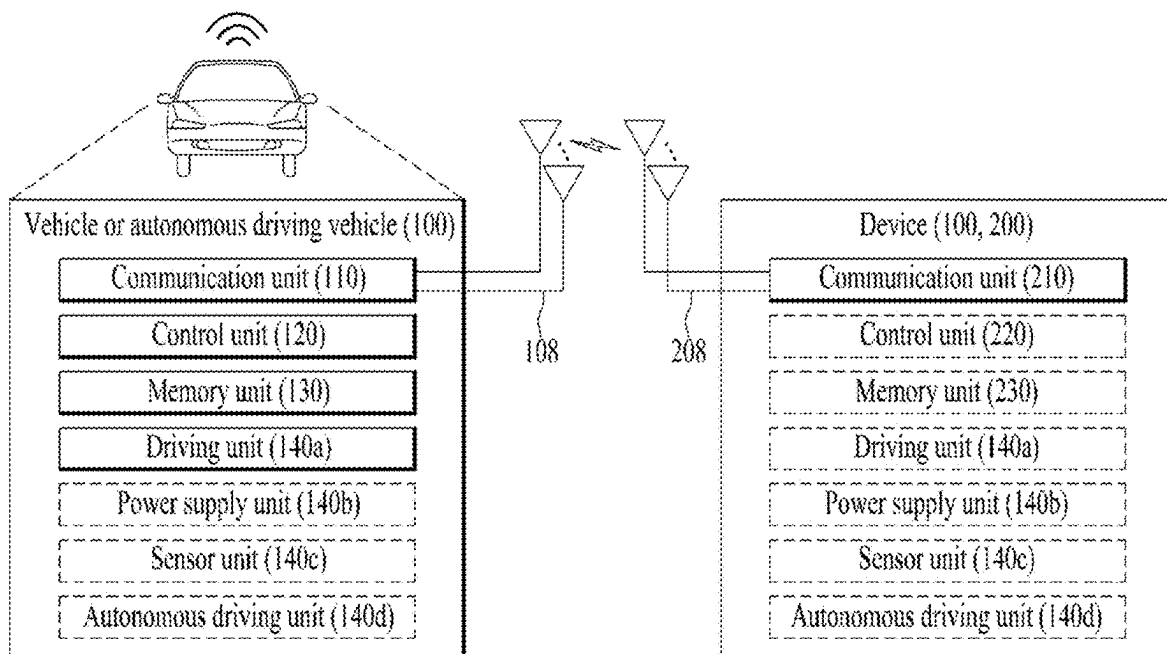


FIG. 25



METHOD BY WHICH TERMINAL CHANGES RECEPTION TIMING IN WIRELESS COMMUNICATION SYSTEM SUPPORTING SIDE LINKS, AND APPARATUS THEREFOR

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is the National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2019/018681, filed on Dec. 30, 2019, the contents of which is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

Embodiments of the present disclosure relate to a method for changing a reception (Rx) timing point of a sidelink signal by a user equipment (UE) in a wireless communication system, and a device for the same.

BACKGROUND ART

Wireless communication systems have been widely deployed to provide various types of communication services such as voice or data. In general, a wireless communication system is a multiple access system that supports communication of multiple users by sharing available system resources (a bandwidth, transmission power, etc.). Examples of multiple access systems include a code division multiple access (CDMA) system, a frequency division multiple access (FDMA) system, a time division multiple access (TDMA) system, an orthogonal frequency division multiple access (OFDMA) system, a single carrier frequency division multiple access (SC-FDMA) system, and a multi carrier frequency division multiple access (MC-FDMA) system.

A sidelink (SL) refers to a communication method in which a direct link is established between user equipment (UE), and voice or data is directly exchanged between terminals without going through a base station (BS). SL is being considered as one way to solve the burden of the base station due to the rapidly increasing data traffic.

V2X (vehicle-to-everything) refers to a communication technology that exchanges information with other vehicles, pedestrians, and infrastructure-built objects through wired/wireless communication. V2X can be divided into four types: vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), vehicle-to-network (V2N), and vehicle-to-pedestrian (V2P). V2X communication may be provided through a PC5 interface and/or a Uu interface.

As more and more communication devices require larger communication capacities in transmitting and receiving signals, there is a need for mobile broadband communication improved from the legacy radio access technology. Accordingly, communication systems considering services/UEs sensitive to reliability and latency are under discussion. A next-generation radio access technology in consideration of enhanced mobile broadband communication, massive Machine Type Communication (MTC), and Ultra-Reliable and Low Latency Communication (URLLC) may be referred to as new radio access technology (RAT) or new radio (NR). Even in NR, vehicle-to-everything (V2X) communication may be supported.

FIG. 1 is a diagram comparing RAT-based V2X communication before NR with NR-based V2X communication.

Regarding V2X communication, in RAT prior to NR, a scheme for providing a safety service based on V2X messages such as a basic safety message (BSM), a cooperative

awareness message (CAM), and a decentralized environmental notification message (DENM) was mainly discussed. The V2X message may include location information, dynamic information, and attribute information. For example, the UE may transmit a periodic message type CAM and/or an event triggered message type DENM to another UE.

For example, the CAM may include dynamic state information about a vehicle such as direction and speed, vehicle static data such as dimensions, and basic vehicle information such as external lighting conditions and route details. For example, a UE may broadcast the CAM, and the CAM latency may be less than 100 ms. For example, when an unexpected situation such as a breakdown of the vehicle or an accident occurs, the UE may generate a DENM and transmit the same to another UE. For example, all vehicles within the transmission coverage of the UE may receive the CAM and/or DENM. In this case, the DENM may have a higher priority than the CAM.

Regarding V2X communication, various V2X scenarios have been subsequently introduced in NR. For example, the various V2X scenarios may include vehicle platooning, advanced driving, extended sensors, and remote driving.

For example, based on vehicle platooning, vehicles may dynamically form a group and move together. For example, to perform platoon operations based on vehicle platooning, vehicles belonging to the group may receive periodic data from a leading vehicle. For example, the vehicles belonging to the group may reduce or increase the distance between the vehicles based on the periodic data.

For example, based on advanced driving, a vehicle may be semi-automated or fully automated. For example, each vehicle may adjust trajectories or maneuvers based on data acquired from local sensors of nearby vehicles and/or nearby logical entities. Also, for example, each vehicle may share driving intention with nearby vehicles.

For example, on the basis of extended sensors, raw data or processed data acquired through local sensors, or live video data may be exchanged between a vehicle, a logical entity, UEs of pedestrians and/or a V2X application server. Thus, for example, the vehicle may recognize an environment that is improved over an environment that may be detected using its own sensor.

For example, for a person who cannot drive or a remote vehicle located in a dangerous environment, a remote driver or V2X application may operate or control the remote vehicle based on remote driving. For example, when a route is predictable as in the case of public transportation, cloud computing-based driving may be used to operate or control the remote vehicle. For example, access to a cloud-based back-end service platform may be considered for remote driving.

A method to specify service requirements for various V2X scenarios such as vehicle platooning, advanced driving, extended sensors, and remote driving is being discussed in the NR-based V2X communication field.

DISCLOSURE

Technical Problem

An object of the present disclosure is to provide a method for adjusting transmission/reception (Tx/Rx) timing points of sidelink to minimize or prevent the influence of inter-symbol interference (ISI) or inter-carrier interference (ICI) between independent sidelinks.

It will be appreciated by persons skilled in the art that the objects that could be achieved with the various embodiments of the present disclosure are not limited to what has been particularly described hereinabove and the above and other objects that the various embodiments of the present disclosure could achieve will be more clearly understood from the following detailed description.

Technical Solutions

In accordance with one aspect of the present disclosure, a method for changing a reception timing point by a first user equipment (UE) in a wireless communication system supporting sidelink may include forming a first sidelink together with a second user equipment (UE) in a first frequency band; estimating a second reception timing point related to a second sidelink formed in a second frequency band, based on a synchronization signal and timing advance (TA) information for a third user equipment (UE) configured to use the second frequency band adjacent to the first frequency band; and determining whether to change a reception timing point of a sidelink signal for the first sidelink, based on a difference value between a first reception timing point related to the first sidelink and the second reception timing point.

The method may further include transmitting, to the second UE, a signal requesting a change in a timing advance (TA) value for the first sidelink based on a change in the reception timing of the sidelink signal.

The reception timing point of the sidelink signal may be changed when the calculated difference value is greater than or equal to a predetermined threshold.

The predetermined threshold may be determined based on a cyclic prefix (CP) length of the first sidelink signal.

The reception timing point of the sidelink signal may be changed when a timing advance (TA) value included in the TA information is greater than the TA value related to the first sidelink.

The reception timing point of the sidelink signal may be changed, when the calculated difference value is greater than or equal to the predetermined threshold and a reception strength of a signal related to the second sidelink is greater than or equal to a predetermined threshold strength.

The reception strength may be a received signal strength indicator (RSSI) or a reference signal received power (RSRP).

The reception timing point of the sidelink signal may be changed to a first timing point determined based on the second reception timing point.

The first timing point may be equal to the second reception timing point.

When an identifier (ID) change of a user equipment (UE) configured to use the second frequency band is detected, the reception timing point of the sidelink signal may be changed from the first timing point to a first reception timing point.

When a reception strength of a signal related to the second sidelink is less than a predetermined threshold strength after the reception timing of the sidelink signal has been changed, the reception timing point of the sidelink signal may be changed from the first timing point to a first reception timing point.

When a difference value between the first reception timing point and a reception timing point at which a signal related to the second sidelink is received is less than a predetermined threshold, a reception timing point of the sidelink signal may be changed to the first reception timing point.

In accordance with another aspect of the present disclosure, a first device configured to transmit a sidelink signal in a wireless communication system supporting sidelink may include: a radio frequency (RF) transceiver; and a processor connected to the RF transceiver, wherein the processor includes forming a first sidelink together with a second device in a first frequency band, under control of the RF transceiver; estimating a second reception timing point related to a second sidelink formed in a second frequency band, based on a synchronization signal and timing advance (TA) information for a third device configured to use the second frequency band adjacent to the first frequency band; and determining whether to change a reception timing point of a sidelink signal for the first sidelink, based on a difference value between a first reception timing point related to the first sidelink and the second reception timing point.

The processor may adjust a driving mode for the device based on the calculated difference value.

Advantageous Effects

Various embodiments of the present disclosure may provide a method for minimizing or preventing the influence of inter-symbol interference (ISI) or inter-carrier interference (ICI) between independent sidelinks by adjusting transmission/reception (Tx/Rx) timing points of sidelink.

Effects to be achieved by embodiment(s) are not limited to what has been particularly described hereinabove and other effects not mentioned herein will be more clearly understood by persons skilled in the art to which embodiment(s) pertain from the following detailed description.

DESCRIPTION OF DRAWINGS

The accompanying drawings, which are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this application, illustrate embodiments of the disclosure and together with the description serve to explain the principle of the disclosure.

FIG. 1 is a diagram for explaining by comparing V2X communication based on RAT before NR and V2X communication based on NR.

FIG. 2 illustrates the structure of an LTE system to which embodiment(s) are applicable.

FIG. 3 illustrates a user-plane radio protocol architecture to which embodiment(s) are applicable.

FIG. 4 illustrates a control-plane radio protocol architecture to which embodiment(s) are applicable.

FIG. 5 illustrates the structure of an NR system to which embodiment(s) are applicable.

FIG. 6 illustrates functional split between an NG-RAN and a 5GC to which embodiment(s) are applicable.

FIG. 7 illustrates the structure of an NR radio frame to which embodiment(s) are applicable.

FIG. 8 illustrates the slot structure of an NR frame to which embodiment(s) are applicable.

FIG. 9 illustrates a radio protocol architecture for SL communication.

FIG. 10 shows the structures of an S-SSB according to CP types.

FIG. 11 illustrates UEs performing V2X or SL communication.

FIG. 12 illustrates resource units for V2X or SL communication.

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FIG. 13 illustrates a procedure in which UEs perform V2X or SL communication according to a transmission mode.

FIG. 14 illustrates a V2X synchronization source or synchronization reference to which embodiments(s) are applicable.

FIG. 15 is a diagram illustrating a case where ISI and ICI occur between independent sidelinks.

FIG. 16 is a diagram illustrating a method for allowing UEs to minimize ISI and ICI caused by an interference signal in sidelink communication.

FIG. 17 is a diagram illustrating a method for determining whether ISI and ICI occur between sidelink signals of UEs.

FIG. 18 is a diagram illustrating an operation for changing a reception (Rx) timing point by a first reception UE.

FIG. 19 is a diagram illustrating an operation for changing a reception timing point by a second reception UE.

FIG. 20 is a diagram illustrating that adjustment of transmission and reception (Tx/Rx) timing points for one sidelink affects another sidelink.

FIG. 21 illustrates a communication system applied to the present disclosure.

FIG. 22 illustrates wireless devices applicable to the present disclosure.

FIG. 23 illustrates another example of a wireless device to which the present disclosure is applied. The wireless device may be implemented in various forms according to use-examples/services.

FIG. 24 illustrates a hand-held device applied to the present disclosure.

FIG. 25 illustrates a vehicle or an autonomous driving vehicle applied to the present disclosure.

BEST MODE

The wireless communication system is a multiple access system that supports communication with multiple users by sharing available system resources (eg, bandwidth, transmission power, etc.). Examples of the multiple access system include a code division multiple access (CDMA) system, a frequency division multiple access (FDMA) system, a time division multiple access (TDMA) system, an orthogonal frequency division multiple access (OFDMA) system, a single carrier frequency (SC-FDMA) system, a multi carrier frequency division multiple access (MC-FDMA) system, and the like.

A sidelink refers to a communication scheme in which a direct link is established between user equipments (UEs) to directly exchange voice or data between UEs without assistance from a base station (BS). The sidelink is being considered as one way to address the burden on the BS caused by rapidly increasing data traffic.

Vehicle-to-everything (V2X) refers to a communication technology for exchanging information with other vehicles, pedestrians, and infrastructure-built objects through wired/wireless communication. V2X may be divided into four types: vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), vehicle-to-network (V2N), and vehicle-to-pedestrian (V2P). V2X communication may be provided through a PC5 interface and/or a Uu interface.

As more and more communication devices require larger communication capacities in transmitting and receiving signals, there is a need for mobile broadband communication improved from the legacy radio access technology. Accordingly, communication systems considering services/UEs sensitive to reliability and latency are under discussion. A next-generation radio access technology in consideration of

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enhanced mobile broadband communication, massive MTC, and Ultra-Reliable and Low Latency Communication (URLLC) may be referred to as new radio access technology (RAT) or new radio (NR). Even in NR, V2X communication may be supported.

Techniques described herein may be used in various wireless access systems such as code division multiple access (CDMA), frequency division multiple access (FDMA), time division multiple access (TDMA), orthogonal frequency division multiple access (OFDMA), single carrier-frequency division multiple access (SC-FDMA), etc. CDMA may be implemented as a radio technology such as universal terrestrial radio access (UTRA) or CDMA2000. TDMA may be implemented as a radio technology such as global system for mobile communications (GSM)/general packet radio service (GPRS)/Enhanced Data Rates for GSM Evolution (EDGE). OFDMA may be implemented as a radio technology such as IEEE 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, evolved-UTRA (E-UTRA) etc. UTRA is a part of universal mobile telecommunications system (UMTS). 3GPP LTE is a part of Evolved UMTS (E-UMTS) using E-UTRA. 3GPP LTE employs OFDMA for downlink and SC-FDMA for uplink. LTE-A is an evolution of 3GPP LTE. 3GPP NR (New Radio or New Radio Access Technology) is an evolved version of 3GPP LTE/LTE-A/LTE-A pro.

5G NR is a successor technology of LTE-A, and is a new clean-slate mobile communication system with characteristics such as high performance, low latency, and high availability. 5G NR can utilize all available spectrum resources, from low frequency bands below 1 GHz to intermediate frequency bands from 1 GHz to 10 GHz and high frequency (millimeter wave) bands above 24 GHz.

For clarity of explanation, LTE-A or 5G NR is mainly described, but the technical spirit of the embodiment(s) is not limited thereto.

FIG. 2 illustrates the structure of an LTE system to which the present disclosure is applicable. This may also be called an evolved UMTS terrestrial radio access network (E-UTRAN) or LTE/LTE-A system.

Referring to FIG. 2, the E-UTRAN includes evolved Node Bs (eNBs) 20 which provide a control plane and a user plane to UEs 10. A UE 10 may be fixed or mobile, and may also be referred to as a mobile station (MS), user terminal (UT), subscriber station (SS), mobile terminal (MT), or wireless device. An eNB 20 is a fixed station communication with the UE 10 and may also be referred to as a base station (BS), a base transceiver system (BTS), or an access point.

eNBs 20 may be connected to each other via an X2 interface. An eNB 20 is connected to an evolved packet core (EPC) 30 via an S1 interface. More specifically, the eNB 20 is connected to a mobility management entity (MME) via an S1-MME interface and to a serving gateway (S-GW) via an S1-U interface.

The EPC 30 includes an MME, an S-GW, and a packet data network-gateway (P-GW). The MME has access information or capability information about UEs, which are mainly used for mobility management of the UEs. The S-GW is a gateway having the E-UTRAN as an end point, and the P-GW is a gateway having a packet data network (PDN) as an end point.

Based on the lowest three layers of the open system interconnection (OSI) reference model known in communication systems, the radio protocol stack between a UE and a network may be divided into Layer 1 (L1), Layer 2 (L2) and Layer 3 (L3). These layers are defined in pairs between a UE and an Evolved UTRAN (E-UTRAN), for data trans-

mission via the Uu interface. The physical (PHY) layer at L1 provides an information transfer service on physical channels. The radio resource control (RRC) layer at L3 functions to control radio resources between the UE and the network. For this purpose, the RRC layer exchanges RRC messages between the UE and an eNB.

FIG. 3 illustrates a user-plane radio protocol architecture to which the present disclosure is applicable.

FIG. 4 illustrates a control-plane radio protocol architecture to which the present disclosure is applicable. A user plane is a protocol stack for user data transmission, and a control plane is a protocol stack for control signal transmission.

Referring to FIGS. 3 and 4, the PHY layer provides an information transfer service to its higher layer on physical channels. The PHY layer is connected to the medium access control (MAC) layer through transport channels and data is transferred between the MAC layer and the PHY layer on the transport channels. The transport channels are divided according to features with which data is transmitted via a radio interface.

Data is transmitted on physical channels between different PHY layers, that is, the PHY layers of a transmitter and a receiver. The physical channels may be modulated in orthogonal frequency division multiplexing (OFDM) and use time and frequencies as radio resources.

The MAC layer provides services to a higher layer, radio link control (RLC) on logical channels. The MAC layer provides a function of mapping from a plurality of logical channels to a plurality of transport channels. Further, the MAC layer provides a logical channel multiplexing function by mapping a plurality of logical channels to a single transport channel. A MAC sublayer provides a data transmission service on the logical channels.

The RLC layer performs concatenation, segmentation, and reassembly for RLC serving data units (SDUs). In order to guarantee various quality of service (QoS) requirements of each radio bearer (RB), the RLC layer provides three operation modes, transparent mode (TM), unacknowledged mode (UM), and acknowledged Mode (AM). An AM RLC provides error correction through automatic repeat request (ARQ).

The RRC layer is defined only in the control plane and controls logical channels, transport channels, and physical channels in relation to configuration, reconfiguration, and release of RBs. An RB refers to a logical path provided by L1 (the PHY layer) and L2 (the MAC layer, the RLC layer, and the packet data convergence protocol (PDCP) layer), for data transmission between the UE and the network.

The user-plane functions of the PDCP layer include user data transmission, header compression, and ciphering. The control-plane functions of the PDCP layer include control-plane data transmission and ciphering/integrity protection.

RB establishment amounts to a process of defining radio protocol layers and channel features and configuring specific parameters and operation methods in order to provide a specific service. RBs may be classified into two types, signaling radio bearer (SRB) and data radio bearer (DRB). The SRB is used as a path in which an RRC message is transmitted on the control plane, whereas the DRB is used as a path in which user data is transmitted on the user plane.

Once an RRC connection is established between the RRC layer of the UE and the RRC layer of the E-UTRAN, the UE is placed in RRC_CONNECTED state, and otherwise, the UE is placed in RRC_IDLE state. In NR, RRC_INACTIVE state is additionally defined. A UE in the RRC_INACTIVE

state may maintain a connection to a core network, while releasing a connection from an eNB.

DL transport channels carrying data from the network to the UE include a broadcast channel (BCH) on which system information is transmitted and a DL shared channel (DL SCH) on which user traffic or a control message is transmitted. Traffic or a control message of a DL multicast or broadcast service may be transmitted on the DL-SCH or a DL multicast channel (DL MCH). UL transport channels carrying data from the UE to the network include a random access channel (RACH) on which an initial control message is transmitted and an UL shared channel (UL SCH) on which user traffic or a control message is transmitted.

The logical channels which are above and mapped to the transport channels include a broadcast control channel (BCCH), a paging control channel (PCCH), a common control channel (CCCH), a multicast control channel (MCCH), and a multicast traffic channel (MTCH).

A physical channel includes a plurality of OFDM symbols in the time domain by a plurality of subcarriers in the frequency domain. One subframe includes a plurality of OFDM symbols in the time domain. An RB is a resource allocation unit defined by a plurality of OFDM symbols by a plurality of subcarriers. Further, each subframe may use specific subcarriers of specific OFDM symbols (e.g., the first OFDM symbol) in a corresponding subframe for a physical DL control channel (PDCCH), that is, an L1/L2 control channel. A transmission time interval (TTI) is a unit time for subframe transmission.

FIG. 5 illustrates the structure of a NR system to which the present disclosure is applicable.

Referring to FIG. 5, a next generation radio access network (NG-RAN) may include a next generation Node B (gNB) and/or an eNB, which provides user-plane and control-plane protocol termination to a UE. In FIG. 5, the NG-RAN is shown as including only gNBs, by way of example. A gNB and an eNB are connected to each other via an Xn interface. The gNB and the eNB are connected to a 5G core network (5GC) via an NG interface. More specifically, the gNB and the eNB are connected to an access and mobility management function (AMF) via an NG-C interface and to a user plane function (UPF) via an NG-U interface.

FIG. 6 illustrates functional split between the NG-RAN and the 5GC to which the present disclosure is applicable.

Referring to FIG. 6, a gNB may provide functions including inter-cell radio resource management (RRM), radio admission control, measurement configuration and provision, and dynamic resource allocation. The AMF may provide functions such as non-access stratum (NAS) security and idle-state mobility processing. The UPF may provide functions including mobility anchoring and protocol data unit (PDU) processing. A session management function (SMF) may provide functions including UE Internet protocol (IP) address allocation and PDU session control.

FIG. 7 illustrates the structure of a NR radio frame to which the present disclosure is applicable.

Referring to FIG. 7, a radio frame may be used for UL transmission and DL transmission in NR. A radio frame is 10 ms in length, and may be defined by two 5-ms half-frames. An HF may include five 1-ms subframes. A subframe may be divided into one or more slots, and the number of slots in an SF may be determined according to a subcarrier spacing (SCS). Each slot may include 12 or 14 OFDM(A) symbols according to a cyclic prefix (CP).

In a normal CP (NCP) case, each slot may include 14 symbols, whereas in an extended CP (ECP) case, each slot

may include 12 symbols. Herein, a symbol may be an OFDM symbol (or CP-OFDM symbol) or an SC-FDMA symbol (or DFT-s-OFDM symbol).

Table 1 below lists the number of symbols per slot $N_{\text{slot_symb}}$, the number of slots per frame N_{frame} , u_{slot} , and the number of slots per subframe N_{subframe} , u_{slot} according to an SCS configuration μ in the NCP case.

TABLE 1

SCS ($15 \cdot 2^u$)	$N_{\text{slot_symb}}$	$N_{\text{frame}}, u_{\text{slot}}$	$N_{\text{subframe}}, u_{\text{slot}}$
15 KHz ($u = 0$)	14	10	1
30 KHz ($u = 1$)	14	20	2
60 KHz ($u = 2$)	14	40	4
120 KHz ($u = 3$)	14	80	8
240 KHz ($u = 4$)	14	160	16

Table 2 below lists the number of symbols per slot, the number of slots per frame, and the number of slots per subframe according to an SCS in the ECP case.

TABLE 2

SCS ($15 \cdot 2^u$)	$N_{\text{slot_symb}}$	$N_{\text{frame}}, u_{\text{slot}}$	$N_{\text{subframe}}, u_{\text{slot}}$
60 KHz ($u = 2$)	12	40	4

In the NR system, different OFDM(A) numerologies (e.g., SCSs, CP lengths, etc.) may be configured for a plurality of cells aggregated for one UE. Thus, the (absolute) duration of a time resource (e.g., SF, slot, or TTI) including the same number of symbols may differ between the aggregated cells (such a time resource is commonly referred to as a time unit (TU) for convenience of description).

In NR, multiple numerologies or SCSs to support various 5G services may be supported. For example, a wide area in conventional cellular bands may be supported when the SCS is 15 kHz, and a dense urban environment, lower latency, and a wider carrier bandwidth may be supported when the SCS is 30 kHz/60 kHz. When the SCS is 60 kHz or higher, a bandwidth wider than 24.25 GHz may be supported to overcome phase noise.

The NR frequency band may be defined as two types of frequency ranges. The two types of frequency ranges may be FR1 and FR2. The numerical values of the frequency ranges may be changed. For example, the two types of frequency ranges may be configured as shown in Table 3 below. Among the frequency ranges used in the NR system, FR1 may represent “sub 6 GHz range” and FR2 may represent “above 6 GHz range” and may be called millimeter wave (mmW).

TABLE 3

Frequency Range designation	Corresponding frequency range	Subcarrier Spacing (SCS)
FR1	450 MHz-6000 MHz	15, 30, 60 kHz
FR2	24250 MHz-52600 MHz	60, 120, 240 kHz

As mentioned above, the numerical values of the frequency ranges of the NR system may be changed. For example, FR1 may include a band of 410 MHz to 7125 MHz as shown in Table 4 below. That is, FR1 may include a frequency band of 6 GHz (or 5850 MHz, 5900 MHz, 5925 MHz, etc.) or higher. For example, the frequency band of 6 GHz (or 5850 MHz, 5900 MHz, 5925 MHz, etc.) or higher included in FR1 may include an unlicensed band. The

unlicensed band may be used for various purposes, for example, for communication for vehicles (e.g., autonomous driving).

TABLE 4

Frequency Range designation	Corresponding frequency range	Subcarrier Spacing (SCS)
FR1	410 MHz-7125 MHz	15, 30, 60 kHz
FR2	24250 MHz-52600 MHz	60, 120, 240 kHz

FIG. 8 illustrates the slot structure of a NR frame to which the present disclosure is applicable.

Referring to FIG. 8, one slot includes a plurality of symbols in the time domain. For example, one slot may include 14 symbols in a normal CP and 12 symbols in an extended CP. Alternatively, one slot may include 7 symbols in the normal CP and 6 symbols in the extended CP.

A carrier may include a plurality of subcarriers in the frequency domain. A resource block (RB) is defined as a plurality of consecutive subcarriers (e.g., 12 subcarriers) in the frequency domain. A bandwidth part (BWP) may be defined as a plurality of consecutive (P)RBs in the frequency domain, and the BWP may correspond to one numerology (e.g., SCS, CP length, etc.). The carrier may include up to N (e.g., 5) BWPs. Data communication may be conducted in an activated BWP. In a resource grid, each element may be referred to as a resource element (RE), and one complex symbol may be mapped thereto.

The wireless interface between UEs or the wireless interface between a UE and a network may be composed of an L1 layer, an L2 layer, and an L3 layer. In various embodiments of the present disclosure, the L1 layer may represent a physical layer. The L2 layer may represent, for example, at least one of a MAC layer, an RLC layer, a PDCP layer, and an SDAP layer. The L3 layer may represent, for example, an RRC layer.

Hereinafter, V2X or sidelink (SL) communication will be described.

FIG. 9 illustrates a radio protocol architecture for SL communication. Specifically, FIG. 9-(a) shows a user plane protocol stack of NR, and FIG. 9-(b) shows a control plane protocol stack of NR.

Hereinafter, a sidelink synchronization signal (SLSS) and synchronization information will be described.

The SLSS is an SL-specific sequence, and may include a primary sidelink synchronization signal (PSSS) and a secondary sidelink synchronization signal (SSSS). The PSSS may be referred to as a sidelink primary synchronization signal (S-PSS), and the SSSS may be referred to as a sidelink secondary synchronization signal (S-SSS). For example, length-127 M-sequences may be used for the S-PSS, and length-127 gold sequences may be used for the S-SSS. For example, the UE may detect an initial signal and acquire synchronization using the S-PSS. For example, the UE may acquire detailed synchronization using the S-PSS and the S-SSS, and may detect a synchronization signal ID.

A physical sidelink broadcast channel (PSBCH) may be a (broadcast) channel on which basic (system) information that the UE needs to know first before transmission and reception of an SL signal is transmitted. For example, the basic information may include SLSS related information, a duplex mode (DM), time division duplex uplink/downlink (TDD UL/DL) configuration, resource pool related information, the type of an application related to the SLSS, a subframe offset, and broadcast information. For example,

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for evaluation of PSBCH performance, the payload size of PSBCH in NR V2X may be 56 bits including CRC of 24 bits.

The S-PSS, S-SSS, and PSBCH may be included in a block format (e.g., an SL synchronization signal (SS)/PSBCH block, hereinafter sidelink-synchronization signal block (S-SSB)) supporting periodic transmission. The S-SSB may have the same numerology (i.e., SCS and CP length) as a physical sidelink control channel (PSSCH)/physical sidelink shared channel (PSSCH) in the carrier, and the transmission bandwidth thereof may be within a (pre)set sidelink BWP (SL BWP). For example, the bandwidth of the S-SSB may be 11 resource blocks (RBs). For example, the PSBCH may span 11 RBs. The frequency position of the S-SSB may be (pre)set. Accordingly, the UE does not need to perform hypothesis detection at a frequency to discover the S-SSB in the carrier.

In the NR SL system, a plurality of numerologies having different SCSs and/or CP lengths may be supported. In this case, as the SCS increases, the length of the time resource in which the transmitting UE transmits the S-SSB may be shortened. Thereby, the coverage of the S-SSB may be narrowed. Accordingly, in order to guarantee the coverage of the S-SSB, the transmitting UE may transmit one or more S-SSBs to the receiving UE within one S-SSB transmission period according to the SCS. For example, the number of S-SSBs that the transmitting UE transmits to the receiving UE within one S-SSB transmission period may be pre-configured or configured for the transmitting UE. For example, the S-SSB transmission period may be 160 ms. For example, for all SCSs, the S-SSB transmission period of 160 ms may be supported.

For example, when the SCS is 15 kHz in FR1, the transmitting UE may transmit one or two S-SSBs to the receiving UE within one S-SSB transmission period. For example, when the SCS is 30 kHz in FR1, the transmitting UE may transmit one or two S-SSBs to the receiving UE within one S-SSB transmission period. For example, when the SCS is 60 kHz in FR1, the transmitting UE may transmit one, two, or four S-SSBs to the receiving UE within one S-SSB transmission period.

For example, when the SCS is 60 kHz in FR2, the transmitting UE may transmit 1, 2, 4, 8, 16 or 32 S-SSBs to the receiving UE within one S-SSB transmission period. For example, when SCS is 120 kHz in FR2, the transmitting UE may transmit 1, 2, 4, 8, 16, 32 or 64 S-SSBs to the receiving UE within one S-SSB transmission period.

When the SCS is 60 kHz, two types of CPs may be supported. In addition, the structure of the S-SSB transmitted from the transmitting UE to the receiving UE may depend on the CP type. For example, the CP type may be normal CP (NCP) or extended CP (ECP). Specifically, for example, when the CP type is NCP, the number of symbols to which the PSBCH is mapped in the S-SSB transmitted by the transmitting UE may be 9 or 8. On the other hand, for example, when the CP type is ECP, the number of symbols to which the PSBCH is mapped in the S-SSB transmitted by the transmitting UE may be 7 or 6. For example, the PSBCH may be mapped to the first symbol in the S-SSB transmitted by the transmitting UE. For example, upon receiving the S-SSB, the receiving UE may perform an automatic gain control (AGC) operation in the period of the first symbol for the S-SSB.

FIG. 10 illustrates the structures of an S-SSB according to CP types. FIG. 10-(a) shows the structure of the S-SSB when the CP type is NCP.

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For example, the structure of the S-SSB, that is, the order of symbols to which the S-PSS, S-SSS, and PSBCH are mapped in the S-SSB transmitted by the transmitting UE when the CP type is NCP may be shown in FIG. 20.

FIG. 10-(b) shows the structure of the S-SSB when the CP type is ECP.

For example, when the CP type is ECP, the number of symbols to which the transmitting UE maps the PSBCH after the S-SSS in the S-SSB may be 6, unlike in FIG. 20. Accordingly, the coverage of the S-SSB may differ between the CP types, NCP and ECP.

Each SLSS may have an SL synchronization identifier (SLSS ID).

For example, in the case of LTE SL or LTE V2X, the value of the SLSS ID may be defined based on a combination of two different S-PSS sequences and 168 different S-SSS sequences. For example, the number of SLSS IDs may be 336. For example, the value of the SLSS ID may be any one of 0 to 335.

For example, in the case of NR SL or NR V2X, the value of the SLSS ID may be defined based on a combination of two different S-PSS sequences and 336 different S-SSS sequences. For example, the number of SLSS IDs may be 672. For example, the value of the SLSS ID may be any one of 0 to 671. For example, one S-PSS of the two different S-PSSs may be associated with in-coverage, and the other S-PSS may be associated with out-of-coverage. For example, SLSS IDs of 0 to 335 may be used in in-coverage, and SLSS IDs of 336 to 671 may be used in out-of-coverage.

In order to improve the S-SSB reception performance of the receiving UE, the transmitting UE needs to optimize the transmit power according to the characteristics of respective signals constituting the S-SSB. For example, according to the peak to average power ratio (PAPR) of each signal constituting the S-SSB, the transmitting UE may determine the value of maximum power reduction (MPR) for each signal. For example, when the PAPR differs between the S-PSS and the S-SSS which constitute the S-SSB, the transmitting UE may apply an optimal MPR value to transmission of each of the S-PSS and the S-SSS in order to improve the S-SSB reception performance of the receiving UE. Also, for example, in order for the transmitting UE to perform an amplification operation on each signal, a transition period may be applied. The transition period may reserve a time required for the transmitter amplifier of the transmitting UE to perform a normal operation at the boundary where the transmit power of the transmitting UE varies. For example, in the case of FR1, the transition period may be 10 μ s. For example, in the case of FR2, the transition period may be 5 μ s. For example, a search window in which the receiving UE is to detect the S-PSS may be 80 ms and/or 160 ms.

FIG. 11 illustrates UEs performing V2X or SL communication.

Referring to FIG. 11, in V2X or SL communication, the term UE may mainly refer to a user's UE. However, when network equipment such as a BS transmits and receives signals according to a communication scheme between UEs, the BS may also be regarded as a kind of UE. For example, UE 1 may be the first device 100, and UE 2 may be the second device 200.

For example, UE 1 may select a resource unit corresponding to a specific resource in a resource pool, which represents a set of resources. Then, UE 1 may transmit an SL signal through the resource unit. For example, UE 2, which is a receiving UE, may receive a configuration of a resource

pool in which UE 1 may transmit a signal, and may detect a signal of UE 1 in the resource pool.

Here, when UE 1 is within the connection range of the BS, the BS may inform UE 1 of a resource pool. On the other hand, when the UE 1 is outside the connection range of the BS, another UE may inform UE 1 of the resource pool, or UE 1 may use a preconfigured resource pool.

In general, the resource pool may be composed of a plurality of resource units, and each UE may select one or multiple resource units and transmit an SL signal through the selected units.

FIG. 12 illustrates resource units for V2X or SL communication.

Referring to FIG. 12, the frequency resources of a resource pool may be divided into NF sets, and the time resources of the resource pool may be divided into NT sets. Accordingly, a total of NF*NT resource units may be defined in the resource pool. FIG. 12 shows an exemplary case where the resource pool is repeated with a periodicity of NT subframes.

As shown in FIG. 12, one resource unit (e.g., Unit #0) may appear periodically and repeatedly. Alternatively, in order to obtain a diversity effect in the time or frequency dimension, an index of a physical resource unit to which one logical resource unit is mapped may change in a predetermined pattern over time. In this structure of resource units, the resource pool may represent a set of resource units available to a UE which intends to transmit an SL signal.

Resource pools may be subdivided into several types. For example, according to the content in the SL signal transmitted in each resource pool, the resource pools may be divided as follows.

(1) Scheduling assignment (SA) may be a signal including information such as a position of a resource through which a transmitting UE transmits an SL data channel, a modulation and coding scheme (MCS) or multiple input multiple output (MIMO) transmission scheme required for demodulation of other data channels, and timing advance (TA). The SA may be multiplexed with SL data and transmitted through the same resource unit. In this case, an SA resource pool may represent a resource pool in which SA is multiplexed with SL data and transmitted. The SA may be referred to as an SL control channel.

(2) SL data channel (physical sidelink shared channel (PSSCH)) may be a resource pool through which the transmitting UE transmits user data. When the SA and SL data are multiplexed and transmitted together in the same resource unit, only the SL data channel except for the SA information may be transmitted in the resource pool for the SL data channel. In other words, resource elements (REs) used to transmit the SA information in individual resource units in the SA resource pool may still be used to transmit the SL data in the resource pool of the SL data channel. For example, the transmitting UE may map the PSSCH to consecutive PRBs and transmit the same.

(3) The discovery channel may be a resource pool used for the transmitting UE to transmit information such as the ID thereof. Through this channel, the transmitting UE may allow a neighboring UE to discover the transmitting UE.

Even when the SL signals described above have the same content, they may use different resource pools according to the transmission/reception properties of the SL signals. For example, even when the SL data channel or discovery message is the same among the signals, it may be classified into different resource pools according to determination of the SL signal transmission timing (e.g., transmission at the reception time of the synchronization reference signal or

transmission by applying a predetermined TA at the reception time), a resource allocation scheme (e.g., the BS designates individual signal transmission resources to individual transmitting UEs or individual transmission UEs select individual signal transmission resources within the resource pool), signal format (e.g., the number of symbols occupied by each SL signal in a subframe, or the number of subframes used for transmission of one SL signal), signal strength from a BS, the strength of transmit power of an SL UE, and the like.

Hereinafter, resource allocation in the SL will be described.

FIG. 13 illustrates a procedure in which UEs perform V2X or SL communication according to a transmission mode. In various embodiments of the present disclosure, the transmission mode may be referred to as a mode or a resource allocation mode. Hereinafter, for simplicity, the transmission mode in LTE may be referred to as an LTE transmission mode, and the transmission mode in NR may be referred to as an NR resource allocation mode.

For example, FIG. 13-(a) illustrates a UE operation related to LTE transmission mode 1 or LTE transmission mode 3. Alternatively, for example, FIG. 24-(a) illustrates a UE operation related to NR resource allocation mode 1. For example, LTE transmission mode 1 may be applied to general SL communication, and LTE transmission mode 3 may be applied to V2X communication.

For example, FIG. 13-(b) illustrates a UE operation related to LTE transmission mode 2 or LTE transmission mode 4. Alternatively, for example, FIG. 24-(b) illustrates a UE operation related to NR resource allocation mode 2.

Referring to FIG. 13-(a), in LTE transmission mode 1, LTE transmission mode 3 or NR resource allocation mode 1, the BS may schedule an SL resource to be used by the UE for SL transmission. For example, the BS may perform resource scheduling for UE 1 through PDCCH (more specifically, downlink control information (DCI)), and UE 1 may perform V2X or SL communication with UE 2 according to the resource scheduling. For example, UE 1 may transmit sidelink control information (SCI) to UE 2 on a physical sidelink control channel (PSCCH), and then transmit data which is based on the SCI to UE 2 on a physical sidelink shared channel (PSSCH).

For example, in NR resource allocation mode 1, the UE may be provided with or allocated resources for one or more SL transmissions of a transport block (TB) from the BS through a dynamic grant. For example, the BS may provide a resource for transmission of the PSCCH and/or PSSCH to the UE using the dynamic grant. For example, the transmitting UE may report the SL hybrid automatic repeat request (HARQ) feedback received from the receiving UE to the BS. In this case, the PUCCH resource and timing for reporting the SL HARQ feedback to the BS may be determined based on an indication in the PDCCH through the BS is to allocate a resource for SL transmission.

For example, DCI may represent a slot offset between DCI reception and a first SL transmission scheduled by DCI. For example, the minimum gap between the DCI scheduling a SL transmission resource and the first scheduled SL transmission resource may not be shorter than the processing time of the corresponding UE.

For example, in NR resource allocation mode 1, the UE may be periodically provided with or allocated a resource set from the BS for a plurality of SL transmissions through a configured grant. For example, the configured grant may include configured grant type 1 or configured grant type 2.

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For example, the UE may determine a TB to be transmitted in each occasion indicated by a given configured grant.

For example, the BS may allocate SL resources to the UE on the same carrier, and may allocate SL resources to the UE on different carriers.

For example, an NR BS may control LTE-based SL communication. For example, the NR BS may transmit NR DCI to the UE to schedule an LTE SL resource. In this case, for example, a new RNTI for scrambling the NR DCI may be defined. For example, the UE may include an NR SL module and an LTE SL module.

For example, after the UE including the NR SL module and the LTE SL module receives NR SL DCI from the gNB, the NR SL module may transform the NR SL DCI to LTE DCI type 5A, and the NR SL module may deliver LTE DCI type 5A to the LTE SL module in units of X ms. For example, the LTE SL module may apply activation and/or release to the first LTE subframe Z ms after the LTE SL module receives LTE DCI format 5A from the NR SL module. For example, the X may be dynamically indicated using a field of DCI. For example, the minimum value of X may depend on the UE capability. For example, the UE may report a single value according to the UE capability. For example, X may be a positive number.

Referring to FIG. 13-(b), in LTE transmission mode 2, LTE transmission mode 4, or NR resource allocation mode 2, the UE may determine AN SL resource within the SL resources configured by the BS/network or the preconfigured SL resources. For example, the configured SL resources or the preconfigured SL resources may be a resource pool. For example, the UE may autonomously select or schedule a resource for SL transmission. For example, the UE may autonomously select a resource within the configured resource pool to perform SL communication. For example, the UE may select a resource within a selection window by performing a sensing and resource (re)selection procedure. For example, the sensing may be performed on a per sub-channel basis. In addition, UE 1, which has selected a resource within the resource pool, may transmit SCI to UE 2 through the PSCCH, and then transmit data, which is based on the SCI, to UE 2 through the PSSCH.

For example, a UE may assist in selecting an SL resource for another UE. For example, in NR resource allocation mode 2, the UE may receive a configured grant for SL transmission. For example, in NR resource allocation mode 2, the UE may schedule SL transmission of another UE. For example, in NR resource allocation mode 2, the UE may reserve an SL resource for blind retransmission.

For example, in NR resource allocation mode 2, UE 1 may indicate the priority of SL transmission to UE 2 using the SCI. For example, UE 2 may decode the SCI. UE 2 may perform sensing and/or resource (re)selection based on the priority. For example, the resource (re)selection procedure may include an operation of identifying candidate resources in a resource selection window by UE 2, and an operation of selecting, by UE 2, a resource for (re)transmission from among the identified candidate resources. For example, the resource selection window may be a time interval during which the UE selects the resource for SL transmission. For example, after UE 2 triggers resource (re)selection, the resource selection window may start at $T_1 \geq 0$. The resource selection window may be limited by the remaining packet delay budget of UE 2. For example, in the operation of identifying the candidate resources in the resource selection window by UE 2, a specific resource may be indicated by the SCI received by UE 2 from UE 1. When the L1 SL RSRP measurement value for the specific resource exceeds an SL

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RSRP threshold, UE 2 may not determine the specific resource as a candidate resource. For example, the SL RSRP threshold may be determined based on the priority of the SL transmission indicated by the SCI received by UE 2 from UE 1 and the priority of the SL transmission on the resource selected by UE 2.

For example, the L1 SL RSRP may be measured based on an SL demodulation reference signal (DMRS). For example, one or more PSSCH DMRS patterns may be configured or preconfigured for each resource pool in the time domain. For example, PDSCH DMRS configuration type 1 and/or type 2 may be the same as or similar to the frequency domain pattern of the PSSCH DMRS. For example, the exact DMRS pattern may be indicated by the SCI. For example, in NR resource allocation mode 2, the transmitting UE may select a specific DMRS pattern from among DMRS patterns configured or preconfigured for the resource pool.

For example, in NR resource allocation mode 2, based on the sensing and resource (re)selection procedure, the transmitting UE may perform initial transmission of a TB without reservation. For example, based on the sensing and resource (re)selection procedure, using the SCI associated with a first TB, the transmitting UE may reserve the SL resource for initial transmission of a second TB.

For example, in NR resource allocation mode 2, the UE may reserve a resource for feedback-based PSSCH retransmission through signaling related to previous transmission of the same TB. For example, the maximum number of SL resources reserved by one transmission including the current transmission may be 2, 3, or 4. For example, the maximum number of SL resources may be the same regardless of whether HARQ feedback is enabled. For example, the maximum number of HARQ (re)transmissions for one TB may be limited by configuration or pre-configuration. For example, the maximum number of HARQ (re)transmissions may be up to 32. For example, when the configuration or pre-configuration is not present, the maximum number of HARQ (re)transmissions may be unspecified. For example, the configuration or pre-configuration may be for the transmitting UE. For example, in NR resource allocation mode 2, HARQ feedback for releasing resources not used by the UE may be supported.

For example, in NR resource allocation mode 2, the UE may indicate to another UE one or more sub-channels and/or slots used by the UE, using the SCI. For example, the UE may indicate to another UE one or more sub-channels and/or slots reserved by the UE for PSSCH (re)transmission, using SCI. For example, the minimum allocation unit of the SL resource may be a slot. For example, the size of the sub-channel may be configured for the UE or may be preconfigured.

Hereinafter, sidelink control information (SCI) will be described.

Control information transmitted by the BS to the UE on the PDCCH may be referred to as downlink control information (DCI), whereas control information transmitted by the UE to another UE on the PSCCH may be referred to as SCI. For example, before decoding the PSCCH, the UE may be aware of the start symbol of the PSCCH and/or the number of symbols of the PSCCH. For example, the SCI may include SL scheduling information. For example, the UE may transmit at least one SCI to another UE to schedule the PSSCH. For example, one or more SCI formats may be defined.

For example, the transmitting UE may transmit the SCI to the receiving UE on the PSCCH. The receiving UE may decode one SCI to receive the PSSCH from the transmitting UE.

For example, the transmitting UE may transmit two consecutive SCIs (e.g., 2-stage SCI) to the receiving UE on the PSCCH and/or the PSSCH. The receiving UE may decode the two consecutive SCIs (e.g., 2-stage SCI) to receive the PSSCH from the transmitting UE. For example, when the SCI configuration fields are divided into two groups in consideration of the (relatively) high SCI payload size, the SCI including a first SCI configuration field group may be referred to as first SCI or 1st SCI, and the SCI including a second SCI configuration field group may be referred to as second SCI or 2nd SCI. For example, the transmitting UE may transmit the first SCI to the receiving UE on the PSCCH. For example, the transmitting UE may transmit the second SCI to the receiving UE on the PSCCH and/or the PSSCH. For example, the second SCI may be transmitted to the receiving UE on the (independent) PSCCH, or may be piggybacked together with data and transmitted on the PSSCH. For example, the two consecutive SCIs may be applied for different transmissions (e.g., unicast, broadcast, or groupcast).

For example, the transmitting UE may transmit some or all of the following information to the receiving UE through SCI. Here, for example, the transmitting UE may transmit some or all of the following information to the receiving UE through the first SCI and/or the second SCI:

- PSSCH and/or PSCCH related resource allocation information, for example, the positions/number of time/frequency resources, resource reservation information (e.g., periodicity); and/or
- SL CSI report request indicator or SL (L1) RSRP (and/or SL (L1) RSRQ and/or SL (L1) RSSI) report request indicator; and/or
- SL CSI transmission indicator (or SL (L1) RSRP (and/or SL (L1) RSRQ and/or SL (L1) RSSI) information transmission indicator) (on PSSCH); and/or
- MCS information; and/or
- transmit power information; and/or
- L1 destination ID information and/or L1 source ID information; and/or
- SL HARQ process ID information; and/or
- new data indicator (NDI) information; and/or
- redundancy version (RV) information; and/or
- (transmission traffic/packet related) QoS information; e.g., priority information; and/or
- SL CSI-RS transmission indicator or information on the number of (transmitted) SL CSI-RS antenna ports;
- Location information about the transmitting UE or location (or distance/area) information about a target receiving UE (to which a request for SL HARQ feedback is made); and/or
- information about a reference signal (e.g., DMRS, etc.) related to decoding and/or channel estimation of data transmitted on the PSSCH, for example, information related to a pattern of a (time-frequency) mapping resource of DMRS, rank information, antenna port index information.

For example, the first SCI may include information related to channel sensing. For example, the receiving UE may decode the second SCI using the PSSCH DMRS. A polar code used for the PDCCH may be applied to the second SCI. For example, in the resource pool, the payload size of the first SCI may be the same for unicast, groupcast and broadcast. After decoding the first SCI, the receiving UE

does not need to perform blind decoding of the second SCI. For example, the first SCI may include scheduling information about the second SCI.

In various embodiments of the present disclosure, since the transmitting UE may transmit at least one of SCI, the first SCI, and/or the second SCI to the receiving UE on the PSCCH, the PSCCH may be replaced/substituted with at least one of the SCI, the first SCI, and/or the second SCI. Additionally/alternatively, for example, the SCI may be replaced/substituted with at least one of the PSCCH, the first SCI, and/or the second SCI. Additionally/alternatively, for example, since the transmitting UE may transmit the second SCI to the receiving UE on the PSSCH, the PSSCH may be replaced/substituted with the second SCI.

Hereinafter, synchronization acquisition by an SL UE will be described.

In TDMA and FDMA systems, accurate time and frequency synchronization is essential. Inaccurate time and frequency synchronization may lead to degradation of system performance due to inter-symbol interference (ISI) and inter-carrier interference (ICI). The same is true for V2X. For time/frequency synchronization in V2X, a sidelink synchronization signal (SLSS) may be used in the PHY layer, and master information block-sidelink-V2X (MIB-SL-V2X) may be used in the RLC layer.

FIG. 14 illustrates a V2X synchronization source or reference to which the present disclosure is applicable.

Referring to FIG. 14, in V2X, a UE may be synchronized with a GNSS directly or indirectly through a UE (within or out of network coverage) directly synchronized with the GNSS. When the GNSS is configured as a synchronization source, the UE may calculate a direct subframe number (DFN) and a subframe number by using a coordinated universal time (UTC) and a (pre)determined DFN offset.

Alternatively, the UE may be synchronized with a BS directly or with another UE which has been time/frequency synchronized with the BS. For example, the BS may be an eNB or a gNB. For example, when the UE is in network coverage, the UE may receive synchronization information provided by the BS and may be directly synchronized with the BS. Thereafter, the UE may provide synchronization information to another neighboring UE. When a BS timing is set as a synchronization reference, the UE may follow a cell associated with a corresponding frequency (when within the cell coverage in the frequency), a primary cell, or a serving cell (when out of cell coverage in the frequency), for synchronization and DL measurement.

The BS (e.g., serving cell) may provide a synchronization configuration for a carrier used for V2X or sidelink communication. In this case, the UE may follow the synchronization configuration received from the BS. When the UE fails in detecting any cell in the carrier used for the V2X or sidelink communication and receiving the synchronization configuration from the serving cell, the UE may follow a predetermined synchronization configuration.

Alternatively, the UE may be synchronized with another UE which has not obtained synchronization information directly or indirectly from the BS or GNSS. A synchronization source and a preference may be preset for the UE. Alternatively, the synchronization source and the preference may be configured for the UE by a control message provided by the BS.

A sidelink synchronization source may be related to a synchronization priority. For example, the relationship between synchronization sources and synchronization priorities may be defined as shown in Tables 5 and 6. Tables 5 and 6 are merely an example, and the relationship between

synchronization sources and synchronization priorities may be defined in various manners.

TABLE 5

Priority	GNSS-based synchronization	BS-based synchronization (eNB/gNB-based synchronization)
P0	GNSS	BS
P1	All UEs directly synchronized with GNSS	All UEs directly synchronized with BS
P2	All UEs indirectly synchronized with GNSS	All UEs indirectly synchronized with BS
P3	All other UEs	GNSS
P4	N/A	All UEs directly synchronized with GNSS
P5	N/A	All UEs indirectly synchronized with GNSS
P6	N/A	All other UEs

TABLE 6

Priority	GNSS-based synchronization	eNB/gNB-based synchronization
P0	GNSS	BS
P1	All UEs directly synchronized with GNSS	All UEs directly synchronized with BS
P2	All UEs indirectly synchronized with GNSS	All UEs indirectly synchronized with BS
P3	BS	GNSS
P4	All UEs directly synchronized with BS	All UEs directly synchronized with GNSS
P5	All UEs indirectly synchronized with BS	All UEs indirectly synchronized with GNSS
P6	Remaining UE(s) with low priority	Remaining UE(s) with low priority

In Table 5 or Table 6, P0 may denote the highest priority, and P6 may denote the lowest priority. In Table 5 or Table 6, the BS may include at least one of a gNB or an eNB.

Whether to use GNSS-based synchronization or BS-based synchronization may be (pre)determined. In a single-carrier operation, the UE may derive its transmission timing from an available synchronization reference with the highest priority.

Transmission Timing Adjustments

In LTE or NR systems, the time it takes for a signal transmitted from the UE to reach the base station (BS) may vary depending on the radius of a cell, the location of the UE in the cell, UE mobility, and the like. That is, when the BS does not control the uplink (UL) transmission (Tx) timing point for each UE, there is a possibility of UE-to-UE interference during communication between the UE and the BS, thereby increasing the error rate at the base station (BS). The time it takes for the signal transmitted from the UE to reach the BS may be referred to as a timing advance (TA). Assuming that the UE is randomly located within the cell, the timing advance (TA) of the UE may vary depending on the UE location. For example, when the UE is located at the boundary of the cell, the timing advance (TA) of the UE may become longer as compared to the other case in which the UE is located at the center of the cell. The timing advance (TA) may vary depending on a frequency band of the cell. Accordingly, in order to prevent interference between the UEs, the BS may have to manage or adjust transmission (Tx) timing points of UEs included in the cell. As described above, the BS operation indicating management or adjustment of the transmission (Tx) timing point may be referred to as maintenance of a timing advance or a time alignment.

Timing advance (TA) maintenance or timing alignment may be performed through the above-described random access procedure. During the random access procedure, the BS may receive a random access preamble, and may calculate a timing advance (TA) value using the received random access preamble. The calculated timing advance (TA) value may be transmitted to the UE through a random access response, and the UE may update the signal transmission timing point based on the received timing advance (TA) value. Alternatively, the BS may calculate the timing advance by receiving an uplink reference signal (e.g., a sounding reference signal (SRS)) transmitted periodically or randomly from the UE, and the UE may calculate the signal transmission (Tx) timing point based on the calculated timing advance (TA) value.

As described above, the BS may measure the timing advance of the UE through the random access preamble or the uplink reference signal, and may inform the UE of an adjustment value for timing alignment. In this case, the adjustment value for timing alignment may be referred to as a timing advance command (TAC). TAC may be processed by the MAC layer. When the UE receives the TAC from the BS, the BS assumes that the received TAC is valid only for a predetermined time. A timing alignment timer (TAT) may be used to indicate the predetermined time. The TAT may be transmitted to the UE through higher layer signaling (e.g., RRC signaling).

Transmission of the uplink radio frame (i) from the UE may start $(NTA + NTA_{\text{offset}}) \times T_s$ (seconds) before the corresponding radio frame begins. In this case, NTA may be denoted by $0 \leq NTA \leq 20512$. In an FDD frame structure, NTA_{offset} may be denoted by 'NTA_{offset}=0'. In a TDD frame structure, NTA_{offset} may be denoted by 'NTA_{offset}=624'. 'NTA' may be instructed by the timing advance command. 'Ts' may represent the sampling time. The uplink transmission (Tx) timing point may be adjusted in units of multiples of 16 Ts. 'TAC' may be given as 11 bits in a random access response and may indicate a value of 0 to 1282. 'NTA' may be given as 'TA*16'. Alternatively, the TAC may be 6 bits and may indicate a value of 0 to 63. In this case, 'NTA' may be given as 'NTA_{old}+(TA-31)*16'. The timing advance command received at the subframe (n) may be applied to subframes starting from the subframe (n+6).

On the other hand, the D2D signal is also transmitted and received using the UL radio resource, which is based on the purpose of mitigating interference problems by maintaining transmission (Tx) resources of the UE. In this case, the D2D signal transmitted by the UE may be broadly divided into two types of D2D signals, i.e., the first-type D2D signal and the second-type D2D signal. In this case, the first-type D2D signal may be a signal to which the existing UL TA is applied without change so that the first-type D2D signal can determine transmission (Tx) resources by direct instruction of the BS (or eNB). The second-type D2D signal may be a signal in which Tx resources are mainly determined by autonomous decision of the UE rather than the direct instruction of the eNB, or may be a signal in which UL TA is not used for smooth multiplexing with other signals to which direct instruction of the BS (or eNB) is not applied, irrespective of the direction instruction of the eNB.

On the other hand, when an individual timing advance (TA) algorithm for a sidelink signal is applied, a time unit boundary may be mismatched between a plurality of independent sidelinks, and thus ISI or ICI may occur between two or more consecutive sidelink signals in a frequency domain. Hereinafter, a method for changing the transmission

and reception (Tx/Rx) timing points of the sidelink based on either the received timing point of the signal transmitted in another sidelink or the signal strength of the signal transmitted in another sidelink will hereinafter be described in detail with reference to the attached drawings.

Method for Avoiding ISI and ICI in Sidelink Communication

A method for avoiding ISI and ICI between independent links in vehicle-to-vehicle (V2V) communication based on beamforming will hereinafter be described. According to the proposed method, during UE-to-UE sidelink communication, a reception (Rx) time between independent links may be calculated to avoid ISI and ICI. Specifically, the arrival time of a signal can be calculated not only using the UE-to-UE distance (or the V2V distance) and the UE-to-UE transmission delay, but also using the TA information applied to data transmission. The V2V distance and the V2V transmission delay may be estimated using a sidelink synchronous signal (SL-Sync signal) transmitted based on a common synchronization reference. The UEs (vehicles) may inform the neighboring UEs of TA information applied to the data signal transmitted in the corresponding beam direction. Alternatively, the proposed method may perform the ISI and ICI avoidance procedure based on the case or condition of ISI/ICI occurrences.

In a situation where the frequency is divided and used for the purpose of reducing interference in sidelink communication, if the reception (Rx) time difference between a plurality of signals transmitted in the respective independent links between UEs (vehicles) becomes greater than or equal to the CP length, ISI and ICI may occur, thereby affecting communication performance. In particular, during vehicle communication in which the positions of a plurality of UEs (a plurality of vehicles) are frequently changed, the above-described degradation of communication performance may become more serious.

For example, if a difference between a reception timing point of a sidelink signal (hereinafter, a target signal) of an independent link to be received by the specific UE and a detection timing point of a sidelink signal (or an interference signal) of an independent link to be received by another adjacent UE is greater than or equal to the CP length, the target signal may generate ISI and ICI due to a difference in reception (Rx) timing between the target signal and the interfering signal. In this case, the specific UE may seriously deteriorate the communication performance for the target signal. Here, the interference signal and the target signal may have frequency bands adjacent to each other in the frequency domain.

That is, if a difference value between the reception (Rx) timing point for a first independent sidelink formed between the specific UE and the first transmission UE and the other reception (Rx) timing point for a second independent sidelink formed between the specific UE and other UEs is equal to or greater than a preset threshold, the sidelink signal generated through the second sidelink may cause ISI/ICI occurrence when the specific UE receives the sidelink signal generated through the first sidelink. A specific problem related thereto will be described with reference to FIG. 15.

FIG. 15 is a diagram illustrating a case where ISI and ICI occur between independent sidelinks.

Referring to FIGS. 15(a) and 15(b), a first reception UE (V0), a first transmission UE (V1), a second reception UE (V3), and a second transmission UE (V2), each of which is a vehicle, may perform sidelink communication. Thus, a first sidelink (that is an independent unicast link between the first reception UE (V0) and the first transmission UE (V1)) and

a second sidelink (that is an independent unicast link between the second reception UE (V3) and the second transmission UE (V2)) are formed. In addition, the first sidelink and the second sidelink may be formed in frequency bands adjacent to each other.

Referring to FIG. 15(a), the first reception UE (V0) may receive a sidelink signal from the first transmission UE (V1) through the first sidelink, and may receive (or sense) a sidelink signal associated with the second sidelink signal as an interference signal. That is, the first reception UE (V0) may receive an undesired signal from the second sidelink. The second reception UE (V3) may receive a desired sidelink signal through the second sidelink, and may receive (or sense) a signal related to the first sidelink signal as an interference signal. In this case, when a difference value between the reception timing point of the interference signal and the reception timing point of the target signal is less than the CP length, the first reception UE (V0) or the second reception UE (V3) may properly receive or decode the target signal without avoiding the ISI/ICI to be generated by the interference signal. In this case, the first reception UE (V0) or the second reception UE (V3) may not perform an operation required for ISI/ICI avoidance.

Referring to FIG. 15(b), the first reception UE (V0) may receive a sidelink signal targeted to the first reception UE through the first sidelink, and may receive an interference signal not targeted to the first reception UE through the second sidelink. In addition, the second reception UE (V3) may receive a sidelink signal targeted to the second reception UE through the second sidelink, and may also receive an interference signal not targeted to the second reception UE through the first sidelink. As shown in FIG. 15(b), in the first reception UE (V0) and the second reception UE (V3), a difference value between the reception timing point for the desired sidelink and the reception timing point for the undesired sidelink may be greater than the CP length. Here, it is difficult to maintain OFDM-related orthogonality between the sidelink signal received through the desired sidelink and the interference signal received through the undesired sidelink in each of the first reception UE (V0) and the second reception UE (V3). In this case, the first reception UE (V0) or the second reception UE (V3) may generate ISI and ICI by the interference signal received from the undesired sidelink. In this case, the first reception UE (V0) or the second reception UE (V3) may perform the operation required for ISI/ICI avoidance.

Therefore, if the difference value is greater than the CP length as shown in FIG. 15(b), the first reception UE (V0) or the second reception UE (V3) should prevent or minimize the ISI/ICI by performing the operation required for ISI/ICI avoidance.

FIG. 16 is a diagram illustrating a method for allowing UEs to minimize ISI and ICI caused by the undesired sidelink signal during sidelink communication.

Referring to FIG. 16, UEs are aligned with a common synchronization reference time of a sidelink synchronization signal (SLSS). That is, the UEs may transmit the sidelink synchronization signal at a timing point aligned based on the timing of the common synchronization reference. The UE having received the sidelink synchronization signal may estimate an STO related to the transmission UE that has transmitted the sidelink synchronization signal based on the timing of the common synchronization reference, and may calculate either the distance to the transmission UE or the transmission delay of the transmission UE based on the

estimated STO. Here, the common synchronization criterion may be a Global Navigation Satellite System (GNSS) architecture, or a gNB.

Next, when transmitting and receiving data, each of the UEs may have a reception time (Rx time or FFT window) aligned with the common synchronization reference. The transmission UE configured to transmit and receive data may apply the timing advance (TA) according to the reception (Rx) time of the vehicle scheduled to receive the signal transmitted by the transmission UE. Further, the transmission UE may broadcast or multicast the TA information applied to a signal to be transmitted.

Specifically, a first sidelink may be formed between the first reception UE (V0) and the first transmission UE (V1), and a second sidelink may be formed between the second reception UE (V3) and the second transmission UE (V2) in a frequency band that is adjacent to the first sidelink while being independent of the first sidelink. The first reception UE (V0) may predict or estimate a second reception timing point at which the undesired interference signal will be received through the second sidelink based on the synchronization signal and TA information transmitted by the second transmission UE. The first reception UE (V0) may calculate a difference value between the second estimated reception (Rx) timing point and the first reception timing point (i.e., a reception (Rx) timing point established in the desired sidelink) for the first sidelink. Based on the calculated difference value, the first reception UE may determine whether ISI or ICI caused by the signal transmitted from the second sidelink occurs during reception of the signal transmitted from the first sidelink. When the occurrence of ISI or ICI in the first sidelink by the second sidelink is predicted, the first reception UE may perform the operation for avoiding the ISI or ICI.

Alternatively, the second reception UE (V3) may predict or estimate a second reception timing point at which the undesired interference signal will be received through the first sidelink based on the synchronization signal and the TA information transmitted by the first transmission UE. The second reception UE (V3) may calculate a difference value between the second estimated reception (Rx) timing point and the first reception timing point (i.e., a reception (Rx) timing point established in the desired sidelink) for the second sidelink. Based on the calculated difference value, the second reception UE (V3) may determine whether ISI or ICI occurs during reception of the second sidelink. When the occurrence of ISI or ICI in the second sidelink by the first sidelink is predicted, the second reception UE (V3) may perform the operation for avoiding the ISI or ICI.

On the other hand, the first reception UE (V0) and the second reception UE (V3) may pre-receive or pre-obtain, from the BS or the neighbor UEs, information on a frequency band used by the neighbor UEs, TA information applied to each of the neighbor UEs, and ID information of the neighbor UEs. In addition, the first reception UE (V0) and the second reception UE (V3) may acquire location information for the neighbor UEs using measurement values of various sensors or GPS information.

FIG. 17 is a diagram illustrating a method for determining whether ISI and ICI occur between sidelink signals of UEs.

Referring to FIG. 17, the reception timing point of the sidelink signal at the first reception UE (V0) and the reception timing point of the sidelink signal at the second reception UE (V3) are aligned based on a common synchronization reference. The first transmission UE (V1) and the first reception UE (V0) may form the first sidelink, and the second transmission UE (V2) and the second reception UE

metric (V3) may form the second sidelink. Here, it is assumed that the frequency band in which the first sidelink is formed and the other frequency band in which the second sidelink is formed are adjacent to each other.

On the other hand, the signal transmitted by the first transmission UE (V1) through the first sidelink may be transmitted not only to a desired first path (Path 10, desired link), but also to an undesired fourth path (Path 20, undesired link). Similarly, the signal transmitted by the second transmission UE (V2) through the second sidelink may be transmitted not only to a desired second path (Path 23, desired link), but also to an undesired third path (Path 13).

In other words, the signal transmitted from the first transmission UE (V1) to the first reception UE (V0) may be transmitted to the undesired second reception UE (V3) through the first path and the fourth path. Similarly, the signal transmitted from the second transmission UE (V2) to the second reception UE (V3) may be transmitted to the undesired first reception UE (V0) through the second path and the third path. In this case, due to the TA value to be applied based on the distance between the second transmission UE (V2) and the second reception UE (V3) as well as a distance difference between the above paths, there may occur a difference between the interference signal and the sidelink signal arriving at the first reception UE (V0) or there may occur a difference between the interference signal and the sidelink signal arriving at the second reception UE (V3). If a difference between the reception timing points is equal to or greater than a preset threshold, ISI and ICI may be generated by the interference signal in the first sidelink signal because orthogonality between the first sidelink signal (or the second sidelink signal) and the interference signal is not maintained.

A difference value ($\Delta 1$) in propagation delay between the first path and the third path and a difference value ($\Delta 2$) in propagation delay between the second path and the fourth path may be greater than the CP length. In this case, the sidelink signal received through the first path in the first reception UE may result in ISI and ICI caused by the signal received through the third path. Also, the sidelink signal received through the second path in the second reception UE may result in ISI and ICI caused by the signal received through the fourth path.

The second reception timing point, which is the reception timing of the interference signal through the second sidelink, may be estimated based on the synchronization signal and the TA information related to the second transmission UE (V2), as described with reference to FIG. 16. That is, when a difference value between a first reception timing point for the first sidelink and a second reception timing point estimated for a signal related to the second sidelink is equal to or greater than a preset threshold, the first reception UE (V0) may change the first reception timing point based on the second reception timing point. In this case, the first reception UE (V0) may transmit, to the first transmission UE (V1), a signal including either information about the changed first reception timing point or a signal requesting the TA change.

For convenience of description, a reception timing point for sidelink desired by the first reception UE (V0) or the second reception UE (V3) may be defined as a first reception timing point, and a reception timing point estimated for a sidelink not desired by the first reception UE (V0) or the second reception UE (V3) may be defined as a second reception timing point. On the other hand, the reception timing point for the undesired sidelink may be estimated based on the synchronization signal and TA information of

the transmission UE configured to transmit a signal on the undesired sidelink, as described with reference to FIG. 15.

According to one embodiment, the first reception UE (V0) may change the reception timing point of the sidelink signal according to a predetermined condition. Specifically, when the difference value between the first reception timing point for the first sidelink and the second reception timing point estimated for the interference signal is equal to or greater than a preset threshold, the change of the first reception timing point may be requested or triggered. Alternatively, if a difference between the first reception timing point for the first sidelink and the estimated second reception timing point is greater than or equal to a preset threshold, and if the reception strength of the received interference signal is equal to or greater than a preset signal strength, the change of the first reception timing point may be requested or triggered. Here, the reception strength may refer to a value of a Received Signal Strength Indicator (RSSI) or a Reference Signal Received Power (RSRP) of the interference signal.

Here, the preset threshold may be predetermined based on the CP length. For example, the preset threshold may be preset to a value corresponding to an extended CP length, or may be preset to a value corresponding to a normal CP length. Alternatively, the preset threshold may be preset to a value determined by the BS. Alternatively, the preset threshold may be preset in consideration of OFDM orthogonality characteristics between a signal transmitted from the desired sidelink and a signal transmitted from the undesired sidelink.

According to one embodiment, the second reception UE (V3) may change the reception timing point of the sidelink signal according to a predetermined condition. Specifically, when the difference value between the first reception timing point for the second sidelink and the second reception timing point estimated for the interference signal is equal to or greater than a preset threshold, the change of the first reception timing point may be requested or triggered. Alternatively, if a difference between the first reception timing point for the second sidelink and the estimated second reception timing point is greater than or equal to a preset threshold, and if the reception strength of the received interference signal is equal to or greater than a preset signal strength, the change of the first reception timing point may be requested or triggered. Here, the reception strength may refer to a value of a Received Signal Strength Indicator (RSSI) or a Reference Signal Received Power (RSRP) of the interference signal.

In addition, when the first reception timing point is changed based on the second reception timing point, the first reception UE (V0) or the second reception UE (V3) may transmit, to the corresponding transmission UE, information on the change of the first reception timing point and the signal requesting the change of the TA value for the corresponding sidelink.

According to one embodiment, only one UE from among the second reception UE (V3) and the first reception UE (V0) may be restricted to perform the reception timing adjustment and the TA change request of the sidelink. That is, only one UE from among the second reception UE (V3) and the first reception UE (V0) may adjust the reception timing point of the sidelink signal based on a difference value between the second reception timing point and the first reception timing point, and may request the TA change for such adjustment of the sidelink signal.

For example, the first reception UE (V0) may determine whether to change the reception timing point of the sidelink

signal based on TA information obtained from the second transmission UE (V2). If the TA value obtained from the second transmission UE (V2) is greater than the TA value applied to the first sidelink, the first reception UE (V0) may adjust the first reception timing point to be aligned with the second reception timing point. The first reception UE (V0) may instruct the first transmission UE (V1) to adjust the TA value corresponding to a change in reception timing for the first sidelink signal, and may inform the BS of the change in reception timing for the first sidelink, so that the BS can instruct the first transmission UE (V1) to change the TA value for the first sidelink. At this time, since the TA value obtained from the first transmission UE (V1) is smaller than the TA value applied to the second sidelink signal, the second reception UE (V3) may not change the reception timing point for the second sidelink.

Alternatively, the first reception UE (V0) may determine whether to change the reception timing point of the first sidelink signal based on ID information of the second transmission UE (V2) and ID information of the first transmission UE (V1). For example, the first reception UE (V0) may change the reception timing point for the first sidelink when the ID of the second transmission UE (V2) has a higher value than the ID of the first transmission UE (V1). In contrast, when the ID of the second transmission UE (V2) has a smaller value than the ID of the first transmission UE (V1), the first reception UE (V0) may change the reception timing point for the first sidelink.

Alternatively, the first reception UE (V0) may change reception resources to be used for reception of the sidelink signal, rather than changing of the reception timing point of the sidelink, based on ID information or TA information of the second transmission UE (V2).

Hereinafter, an exemplary case in which the first reception UE (V0) or the second reception UE (V3) performs the operation of changing the reception timing point for the desired sidelink based on the reception timing point estimated for the desired sidelink will be described in detail.

FIG. 18 is a diagram illustrating an operation for changing the reception (Rx) timing point by the first reception UE. FIG. 19 is a diagram illustrating an operation for changing the reception timing point by the second reception UE.

Here, a first sidelink is formed between the first reception UE (V0) and the first transmission UE (V1) and a second sidelink is formed between the second reception UE (V3) and the second transmission UE (V2), and it is assumed that the first sidelink and the second sidelink are formed in frequency bands adjacent to each other.

Referring to FIG. 18, the reception timing point for the first sidelink at the first reception UE (V0) has been changed. Specifically, the reception timing point (hereinafter, the first reception timing point) of the first reception UE (V0) for the first sidelink is aligned with the reception timing point (hereinafter, the second reception timing point) of the interference signal through the second sidelink. In this case, the reception timing point of the signal (Path 10) received by the first reception UE (V0) in the first sidelink and the interference signal (Path 20) received by the first reception UE V0 in the second sidelink may have a difference value less than the CP length. In addition, the reception timing point between the signal (Path 23) received by the second reception UE (V3) in the second sidelink and the interference signal (Path 13) received by the second reception UE (V3) in the first sidelink may have a difference value less than the CP length. That is, by changing the reception timing point for the first sidelink based on the reception timing point estimated for the interference signal of the first reception

UE, ISI and ICI that may occur in the first reception UE can be prevented. Furthermore, ISI and ICI that may occur in the second reception UE can also be prevented or minimized.

Alternatively, referring to FIG. 19, the reception timing point for the second sidelink at the second reception UE (V3) has been changed. Specifically, the reception timing point (Rx timing) of the second reception UE (V3) for the second sidelink is aligned with the reception timing point of the interference signal through the first sidelink. The reception timing point of the interference signal through the first sidelink may be estimated based on the synchronization signal and the TA information related to the first transmission UE (V1), as described with reference to FIG. 16. That is, the second reception UE (V3) may change the first reception timing point based on the second reception timing point when a difference value between the first reception timing point for the first sidelink and the second reception timing point estimated for the signal related to the second sidelink is equal to or greater than a preset threshold. In this case, the reception timing point between the signal (Path 23) received by the second reception UE (V3) in the second sidelink and the interference signal (Path 13) received by the second reception UE V3 in the first sidelink may have a difference value less than the CP length. Further, the reception timing point between the signal (Path 10) received by the first reception UE (V0) in the first sidelink and the interference signal (Path 20) received by the first reception UE (V0) in the second sidelink may have a difference value less than the CP length. That is, by changing the reception timing point for the second sidelink based on the reception timing point estimated for the interference signal of the second reception UE, ISI and ICI that may occur in the second reception UE can be prevented. Furthermore, ISI and ICI that may occur in the first reception UE can also be prevented or minimized.

According to one embodiment, when the first reception UE (V0) or the second reception UE (V3) satisfies a predetermined condition, the changed reception timing point (hereinafter, the second timing point) can return to the previous reception timing point (hereinafter, the first timing point). Specifically, in a first case where UE ID using the second sidelink is changed, in a second case where the difference value between the second reception timing point and the first reception timing point is less than the preset threshold, in a third case where the difference value between the second reception timing point and the first reception timing point is less than a preset threshold strength of RSSI or RSRP for the signal received in the second sidelink, in a fourth case where new resources for the first sidelink are allocated, or in a fifth case where the first reception UE (V0) receives a return request of the reception timing point from the BS, the reception timing point for the first sidelink can return to the first timing point.

FIG. 20 is a diagram illustrating that adjustment of transmission and reception (Tx/Rx) timing points for one sidelink affects another sidelink.

Referring to FIG. 20(a), a first sidelink (A-C), a second sidelink (B-D), and a third sidelink (E-F) are formed. When the frequency band of the first sidelink and the frequency band of the second sidelink are adjacent to each other, the transmission signal (A) may generate the ISI and may be received at the sidelink (D). In order to prevent ISI occurrence, the second sidelink may change the reception timing point and the transmission timing point. At this time, due to the change in the transmission/reception (Tx/Rx) timing points of the second sidelink, the signal transmitted from the second sidelink may generate ISI in the third sidelink

adjacent to the frequency band of the second sidelink. In this case, similar to the second sidelink, ICI can be avoided by changing the transmission and reception (Tx/Rx) timing points of the third sidelink signal. On the other hand, the effect on the other sidelink in response to the change in the transmission and reception (Tx/Rx) timing points of the sidelink may be problematic only in the dotted region shown in FIG. 20(a). That is, the sidelink for a vehicle G (Vehicle G) deviating from the dotted region may not be affected by the change in the transmission and reception (Tx/Rx) timing points of the second sidelink and the third sidelink when considering the beam direction, the pathloss, and transmission (Tx) power of the vehicle B and the vehicle E.

Referring to FIG. 20(b), the first sidelink (A-C), the second sidelink (B-D), and the third sidelink (E-F) are formed. Here, the third sidelink is formed in a frequency band spaced apart from the second sidelink. When the frequency bands of the first sidelink and the second sidelink are adjacent to each other, the transmission signal (A) may generate ISI and may be received at the sidelink (D). In order to prevent ISI occurrence, the second sidelink may change the reception timing point and the transmission timing point. In this case, unlike FIG. 20(a), since the second sidelink and the third sidelink are formed in frequency bands not adjacent to each other, the third sidelink may not be affected by the change in the transmission and reception (Tx/Rx) timing points of the second sidelink.

Communication System Example to which the Present Invention is Applied

Although not limited thereto, various descriptions, functions, procedures, proposals, methods, and/or operational flow charts of the present invention disclosed in this document can be applied to various fields requiring wireless communication/connection (5G) between devices.

Hereinafter, it will be illustrated in more detail with reference to the drawings. In the following drawings/description, the same reference numerals may exemplify the same or corresponding hardware blocks, software blocks, or functional blocks, unless otherwise indicated.

FIG. 21 illustrates a communication system applied to the present invention.

Referring to FIG. 21, a communication system 1 applied to the present invention includes wireless devices, Base Stations (BSs), and a network. Herein, the wireless devices represent devices performing communication using Radio Access Technology (RAT) (e.g., 5G New RAT (NR)) or Long-Term Evolution (LTE)) and may be referred to as communication/radio/5G devices. The wireless devices may include, without being limited to, a robot 100a, vehicles 100b-1 and 100b-2, an eXtended Reality (XR) device 100c, a hand-held device 100d, a home appliance 100e, an Internet of Things (IoT) device 100f, and an Artificial Intelligence (AI) device/server 400. For example, the vehicles may include a vehicle having a wireless communication function, an autonomous driving vehicle, and a vehicle capable of performing communication between vehicles. Herein, the vehicles may include an Unmanned Aerial Vehicle (UAV) (e.g., a drone). The XR device may include an Augmented Reality (AR)/Virtual Reality (VR)/Mixed Reality (MR) device and may be implemented in the form of a Head-Mounted Device (HMD), a Head-Up Display (HUD) mounted in a vehicle, a television, a smartphone, a computer, a wearable device, a home appliance device, a digital signage, a vehicle, a robot, etc. The hand-held device may include a smartphone, a smartpad, a wearable device (e.g., a smartwatch or a smartglasses), and a computer (e.g., a notebook). The home appliance may include a TV, a refrig-

erator, and a washing machine. The IoT device may include a sensor and a smartmeter. For example, the BSs and the network may be implemented as wireless devices and a specific wireless device 200a may operate as a BS/network node with respect to other wireless devices.

The wireless devices 100a to 100f may be connected to the network 300 via the BSs 200. An AI technology may be applied to the wireless devices 100a to 100f and the wireless devices 100a to 100f may be connected to the AI server 400 via the network 300. The network 300 may be configured using a 3G network, a 4G (e.g., LTE) network, or a 5G (e.g., NR) network. Although the wireless devices 100a to 100f may communicate with each other through the BSs 200/network 300, the wireless devices 100a to 100f may perform direct communication (e.g., sidelink communication) with each other without passing through the BSs/network. For example, the vehicles 100b-1 and 100b-2 may perform direct communication (e.g. Vehicle-to-Vehicle (V2V)/Vehicle-to-everything (V2X) communication). The IoT device (e.g., a sensor) may perform direct communication with other IoT devices (e.g., sensors) or other wireless devices 100a to 100f.

Wireless communication/connections 150a, 150b, or 150c may be established between the wireless devices 100a to 100f/BS 200, or BS 200/BS 200. Herein, the wireless communication/connections may be established through various RATs (e.g., 5G NR) such as uplink/downlink communication 150a, sidelink communication 150b (or, D2D communication), or inter BS communication (e.g. relay, Integrated Access Backhaul (IAB)). The wireless devices and the BSs/the wireless devices may transmit/receive radio signals to/from each other through the wireless communication/connections 150a and 150b. For example, the wireless communication/connections 150a and 150b may transmit/receive signals through various physical channels. To this end, at least a part of various configuration information configuring processes, various signal processing processes (e.g., channel encoding/decoding, modulation/demodulation, and resource mapping/demapping), and resource allocating processes, for transmitting/receiving radio signals, may be performed based on the various proposals of the present invention.

Examples of Wireless Devices to which the Present Invention is Applied

FIG. 22 illustrates a wireless device applicable to the present invention.

Referring to FIG. 22, a first wireless device 100 and a second wireless device 200 may transmit radio signals through a variety of RATs (e.g., LTE and NR). Herein, {the first wireless device 100 and the second wireless device 200} may correspond to {the wireless device 100x and the BS 200} and/or {the wireless device 100x and the wireless device 100x} of FIG. 21.

The first wireless device 100 may include one or more processors 102 and one or more memories 104 and additionally further include one or more transceivers 106 and/or one or more antennas 108. The processor(s) 102 may control the memory(s) 104 and/or the transceiver(s) 106 and may be configured to implement the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. For example, the processor(s) 102 may process information within the memory(s) 104 to generate first information/signals and then transmit radio signals including the first information/signals through the transceiver(s) 106. The processor(s) 102 may receive radio signals including second information/signals through the transceiver 106 and then store information obtained by

processing the second information/signals in the memory(s) 104. The memory(s) 104 may be connected to the processor(s) 102 and may store a variety of information related to operations of the processor(s) 102. For example, the memory(s) 104 may store software code including commands for performing a part or the entirety of processes controlled by the processor(s) 102 or for performing the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. Herein, the processor(s) 102 and the memory(s) 104 may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver(s) 106 may be connected to the processor(s) 102 and transmit and/or receive radio signals through one or more antennas 108. Each of the transceiver(s) 106 may include a transmitter and/or a receiver. The transceiver(s) 106 may be interchangeably used with Radio Frequency (RF) unit(s). In the present invention, the wireless device may represent a communication modem/circuit/chip.

The second wireless device 200 may include one or more processors 202 and one or more memories 204 and additionally further include one or more transceivers 206 and/or one or more antennas 208. The processor(s) 202 may control the memory(s) 204 and/or the transceiver(s) 206 and may be configured to implement the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. For example, the processor(s) 202 may process information within the memory(s) 204 to generate third information/signals and then transmit radio signals including the third information/signals through the transceiver(s) 206. The processor(s) 202 may receive radio signals including fourth information/signals through the transceiver(s) 106 and then store information obtained by processing the fourth information/signals in the memory(s) 204. The memory(s) 204 may be connected to the processor(s) 202 and may store a variety of information related to operations of the processor(s) 202. For example, the memory(s) 204 may store software code including commands for performing a part or the entirety of processes controlled by the processor(s) 202 or for performing the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. Herein, the processor(s) 202 and the memory(s) 204 may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver(s) 206 may be connected to the processor(s) 202 and transmit and/or receive radio signals through one or more antennas 208. Each of the transceiver(s) 206 may include a transmitter and/or a receiver. The transceiver(s) 206 may be interchangeably used with RF unit(s). In the present invention, the wireless device may represent a communication modem/circuit/chip.

Hereinafter, hardware elements of the wireless devices 100 and 200 will be described more specifically. One or more protocol layers may be implemented by, without being limited to, one or more processors 102 and 202. For example, the one or more processors 102 and 202 may implement one or more layers (e.g., functional layers such as PHY, MAC, RLC, PDCP, RRC, and SDAP). The one or more processors 102 and 202 may generate one or more Protocol Data Units (PDUs) and/or one or more Service Data Unit (SDUs) according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. The one or more processors 102 and 202 may generate messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or opera-

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tional flowcharts disclosed in this document. The one or more processors **102** and **202** may generate signals (e.g., baseband signals) including PDUs, SDUs, messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document and provide the generated signals to the one or more transceivers **106** and **206**. The one or more processors **102** and **202** may receive the signals (e.g., baseband signals) from the one or more transceivers **106** and **206** and acquire the PDUs, SDUs, messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document.

The one or more processors **102** and **202** may be referred to as controllers, microcontrollers, microprocessors, or microcomputers. The one or more processors **102** and **202** may be implemented by hardware, firmware, software, or a combination thereof. As an example, one or more Application Specific Integrated Circuits (ASICs), one or more Digital Signal Processors (DSPs), one or more Digital Signal Processing Devices (DSPDs), one or more Programmable Logic Devices (PLDs), or one or more Field Programmable Gate Arrays (FPGAs) may be included in the one or more processors **102** and **202**. The descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be implemented using firmware or software and the firmware or software may be configured to include the modules, procedures, or functions. Firmware or software configured to perform the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be included in the one or more processors **102** and **202** or stored in the one or more memories **104** and **204** so as to be driven by the one or more processors **102** and **202**. The descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be implemented using firmware or software in the form of code, commands, and/or a set of commands.

The one or more memories **104** and **204** may be connected to the one or more processors **102** and **202** and store various types of data, signals, messages, information, programs, code, instructions, and/or commands. The one or more memories **104** and **204** may be configured by Read-Only Memories (ROMs), Random Access Memories (RAMs), Electrically Erasable Programmable Read-Only Memories (EPROMs), flash memories, hard drives, registers, cash memories, computer-readable storage media, and/or combinations thereof. The one or more memories **104** and **204** may be located at the interior and/or exterior of the one or more processors **102** and **202**. The one or more memories **104** and **204** may be connected to the one or more processors **102** and **202** through various technologies such as wired or wireless connection.

The one or more transceivers **106** and **206** may transmit user data, control information, and/or radio signals/channels, mentioned in the methods and/or operational flowcharts of this document, to one or more other devices. The one or more transceivers **106** and **206** may receive user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document, from one or more other devices. For example, the one or more transceivers **106** and **206** may be connected to the one or more processors **102** and **202** and transmit and receive radio signals. For example, the one or more processors **102** and **202** may perform control so that the one or

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more transceivers **106** and **206** may transmit user data, control information, or radio signals to one or more other devices. The one or more processors **102** and **202** may perform control so that the one or more transceivers **106** and **206** may receive user data, control information, or radio signals from one or more other devices. The one or more transceivers **106** and **206** may be connected to the one or more antennas **108** and **208** and the one or more transceivers **106** and **206** may be configured to transmit and receive user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document, through the one or more antennas **108** and **208**. In this document, the one or more antennas may be a plurality of physical antennas or a plurality of logical antennas (e.g., antenna ports). The one or more transceivers **106** and **206** may convert received radio signals/channels etc. from RF band signals into baseband signals in order to process received user data, control information, radio signals/channels, etc. using the one or more processors **102** and **202**. The one or more transceivers **106** and **206** may convert the user data, control information, radio signals/channels, etc. processed using the one or more processors **102** and **202** from the base band signals into the RF band signals. To this end, the one or more transceivers **106** and **206** may include (analog) oscillators and/or filters.

Examples of Wireless Devices to which the Present Invention is Applied

FIG. 23 illustrates another example of a wireless device applied to the present invention. The wireless device may be implemented in various forms according to a use-case/service (refer to FIG. 21)

Referring to FIG. 23, wireless devices **100** and **200** may correspond to the wireless devices **100** and **200** of FIG. 22 and may be configured by various elements, components, units/portions, and/or modules. For example, each of the wireless devices **100** and **200** may include a communication unit **110**, a control unit **120**, a memory unit **130**, and additional components **140**. The communication unit may include a communication circuit **112** and transceiver(s) **114**. For example, the communication circuit **112** may include the one or more processors **102** and **202** and/or the one or more memories **104** and **204** of FIG. 22. For example, the transceiver(s) **114** may include the one or more transceivers **106** and **206** and/or the one or more antennas **108** and **208** of FIG. 22. The control unit **120** is electrically connected to the communication unit **110**, the memory **130**, and the additional components **140** and controls overall operation of the wireless devices. For example, the control unit **120** may control an electric/mechanical operation of the wireless device based on programs/code/commands/information stored in the memory unit **130**. The control unit **120** may transmit the information stored in the memory unit **130** to the exterior (e.g., other communication devices) via the communication unit **110** through a wireless/wired interface or store, in the memory unit **130**, information received through the wireless/wired interface from the exterior (e.g., other communication devices) via the communication unit **110**.

The additional components **140** may be variously configured according to types of wireless devices. For example, the additional components **140** may include at least one of a power unit/battery, input/output (I/O) unit, a driving unit, and a computing unit. The wireless device may be implemented in the form of, without being limited to, the robot (**100a** of FIG. 21), the vehicles (**100b-1** and **100b-2** of FIG. 21), the XR device (**100c** of FIG. 21), the hand-held device

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(100d of FIG. 21), the home appliance (100e of FIG. 21), the IoT device (100f of FIG. 21), a digital broadcast terminal, a hologram device, a public safety device, an MTC device, a medicine device, a fintech device (or a finance device), a security device, a climate/environment device, the AI server/device (400 of FIG. 21), the BSs (200 of FIG. 21), a network node, etc. The wireless device may be used in a mobile or fixed place according to a use-example/service.

In FIG. 23, the entirety of the various elements, components, units/portions, and/or modules in the wireless devices 100 and 200 may be connected to each other through a wired interface or at least a part thereof may be wirelessly connected through the communication unit 110. For example, in each of the wireless devices 100 and 200, the control unit 120 and the communication unit 110 may be connected by wire and the control unit 120 and first units (e.g., 130 and 140) may be wirelessly connected through the communication unit 110. Each element, component, unit/portion, and/or module within the wireless devices 100 and 200 may further include one or more elements. For example, the control unit 120 may be configured by a set of one or more processors. As an example, the control unit 120 may be configured by a set of a communication control processor, an application processor, an Electronic Control Unit (ECU), a graphical processing unit, and a memory control processor. As another example, the memory 130 may be configured by a Random Access Memory (RAM), a Dynamic RAM (DRAM), a Read Only Memory (ROM), a flash memory, a volatile memory, a non-volatile memory, and/or a combination thereof.

Hereinafter, an example of implementing FIG. 23 will be described in detail with reference to the drawings. Examples of Mobile Devices to which the Present Invention is Applied

FIG. 24 illustrates a hand-held device applied to the present invention. The hand-held device may include a smartphone, a smartpad, a wearable device (e.g., a smart-watch or a smartglasses), or a portable computer (e.g., a notebook). The hand-held device may be referred to as a mobile station (MS), a user terminal (UT), a Mobile Subscriber Station (MSS), a Subscriber Station (SS), an Advanced Mobile Station (AMS), or a Wireless Terminal (WT).

Referring to FIG. 24, a hand-held device 100 may include an antenna unit 108, a communication unit 110, a control unit 120, a memory unit 130, a power supply unit 140a, an interface unit 140b, and an I/O unit 140c. The antenna unit 108 may be configured as a part of the communication unit 110. Blocks 110 to 130/140a to 140c correspond to the blocks 110 to 130/140 of FIG. 23, respectively.

The communication unit 110 may transmit and receive signals (e.g., data and control signals) to and from other wireless devices or BSs. The control unit 120 may perform various operations by controlling constituent elements of the hand-held device 100. The control unit 120 may include an Application Processor (AP). The memory unit 130 may store data/parameters/programs/code/commands needed to drive the hand-held device 100. The memory unit 130 may store input/output data/information. The power supply unit 140a may supply power to the hand-held device 100 and include a wired/wireless charging circuit, a battery, etc. The interface unit 140b may support connection of the hand-held device 100 to other external devices. The interface unit 140b may include various ports (e.g., an audio I/O port and a video I/O port) for connection with external devices. The I/O unit 140c may input or output video information/signals, audio information/signals, data, and/or information input by

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a user. The I/O unit 140c may include a camera, a microphone, a user input unit, a display unit 140d, a speaker, and/or a haptic module.

As an example, in the case of data communication, the I/O unit 140c may acquire information/signals (e.g., touch, text, voice, images, or video) input by a user and the acquired information/signals may be stored in the memory unit 130. The communication unit 110 may convert the information/signals stored in the memory into radio signals and transmit the converted radio signals to other wireless devices directly or to a BS. The communication unit 110 may receive radio signals from other wireless devices or the BS and then restore the received radio signals into original information/signals. The restored information/signals may be stored in the memory unit 130 and may be output as various types (e.g., text, voice, images, video, or haptic) through the I/O unit 140c.

Examples of Vehicles or Autonomous Vehicles to which the Present Invention is Applied

FIG. 25 illustrates a vehicle or an autonomous driving vehicle applied to the present invention. The vehicle or autonomous driving vehicle may be implemented by a mobile robot, a car, a train, a manned/unmanned Aerial Vehicle (AV), a ship, etc.

Referring to FIG. 25, a vehicle or autonomous driving vehicle 100 may include an antenna unit 108, a communication unit 110, a control unit 120, a driving unit 140a, a power supply unit 140b, a sensor unit 140c, and an autonomous driving unit 140d. The antenna unit 108 may be configured as a part of the communication unit 110. The blocks 110/130/140a to 140d correspond to the blocks 110/130/140 of FIG. 23, respectively.

The communication unit 110 may transmit and receive signals (e.g., data and control signals) to and from external devices such as other vehicles, BSs (e.g., gNBs and road side units), and servers. The control unit 120 may perform various operations by controlling elements of the vehicle or the autonomous driving vehicle 100. The control unit 120 may include an Electronic Control Unit (ECU). Also, the driving unit 140a may cause the vehicle or the autonomous driving vehicle 100 to drive on a road. The driving unit 140a may include an engine, a motor, a powertrain, a wheel, a brake, a steering device, etc. The power supply unit 140b may supply power to the vehicle or the autonomous driving vehicle 100 and include a wired/wireless charging circuit, a battery, etc. The sensor unit 140c may acquire a vehicle state, ambient environment information, user information, etc. The sensor unit 140c may include an Inertial Measurement Unit (IMU) sensor, a collision sensor, a wheel sensor, a speed sensor, a slope sensor, a weight sensor, a heading sensor, a position module, a vehicle forward/backward sensor, a battery sensor, a fuel sensor, a tire sensor, a steering sensor, a temperature sensor, a humidity sensor, an ultrasonic sensor, an illumination sensor, a pedal position sensor, etc. The autonomous driving unit 140d may implement technology for maintaining a lane on which a vehicle is driving, technology for automatically adjusting speed, such as adaptive cruise control, technology for autonomously driving along a determined path, technology for driving by automatically setting a path if a destination is set, and the like.

For example, the communication unit 110 may receive map data, traffic information data, etc. from an external server. The autonomous driving unit 140d may generate an autonomous driving path and a driving plan from the obtained data. The control unit 120 may control the driving unit 140a such that the vehicle or the autonomous driving

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vehicle **100** may move along the autonomous driving path according to the driving plan (e.g., speed/direction control). In the middle of autonomous driving, the communication unit **110** may aperiodically/periodically acquire recent traffic information data from the external server and acquire surrounding traffic information data from neighboring vehicles. In the middle of autonomous driving, the sensor unit **140c** may obtain a vehicle state and/or surrounding environment information. The autonomous driving unit **140d** may update the autonomous driving path and the driving plan based on the newly obtained data/information. The communication unit **110** may transfer information about a vehicle position, the autonomous driving path, and/or the driving plan to the external server. The external server may predict traffic information data using AI technology, etc., based on the information collected from vehicles or autonomous driving vehicles and provide the predicted traffic information data to the vehicles or the autonomous driving vehicles.

The embodiments described above are those in which components and features of the present invention are combined in a predetermined form. Each component or feature should be considered optional unless explicitly stated otherwise. Each component or feature may be implemented in a form that is not combined with other components or features. In addition, it is also possible to constitute an embodiment of the present invention by combining some components and/or features. The order of operations described in the embodiments of the present invention may be changed. Some configurations or features of one embodiment may be included in other embodiments, or may be replaced with corresponding configurations or features of other embodiments. It is obvious that the embodiments may be configured by combining claims that do not have an explicit citation relationship in the claims or may be included as new claims by amendment after filing.

In this document, embodiments of the present invention have been mainly described based on a signal transmission/reception relationship between a terminal and a base station. Such a transmission/reception relationship is extended in the same/similar manner to signal transmission/reception between a terminal and a relay or a base station and a relay. A specific operation described as being performed by a base station in this document may be performed by its upper node in some cases. That is, it is obvious that various operations performed for communication with a terminal in a network comprising a plurality of network nodes including a base station may be performed by the base station or network nodes other than the base station. The base station may be replaced by terms such as a fixed station, a Node B, an eNode B (eNB), an access point, and the like. In addition, the terminal may be replaced with terms such as User Equipment (UE), Mobile Station (MS), Mobile Subscriber Station (MSS).

In a hardware configuration, the embodiments of the present disclosure may be achieved by one or more application specific integrated circuits (ASICs), digital signal processors (DSPs), digital signal processing devices (DSPDs), programmable logic devices (PLDs), field programmable gate arrays (FPGAs), processors, controllers, microcontrollers, microprocessors, etc.

In a firmware or software configuration, a method according to embodiments of the present disclosure may be implemented in the form of a module, a procedure, a function, etc. Software code may be stored in a memory unit and executed by a processor. The memory unit is located at the interior or exterior of the processor and may transmit and receive data to and from the processor via various known means

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As described before, a detailed description has been given of preferred embodiments of the present disclosure so that those skilled in the art may implement and perform the present disclosure. While reference has been made above to the preferred embodiments of the present disclosure, those skilled in the art will understand that various modifications and alterations may be made to the present disclosure within the scope of the present disclosure. For example, those skilled in the art may use the components described in the foregoing embodiments in combination. The above embodiments are therefore to be construed in all aspects as illustrative and not restrictive. The scope of the disclosure should be determined by the appended claims and their legal equivalents, not by the above description, and all changes coming within the meaning and equivalency range of the appended claims are intended to be embraced therein.

INDUSTRIAL APPLICABILITY

The above-described embodiments of the present disclosure are applicable to various mobile communication systems.

What is claimed is:

1. A method by a first user equipment (UE) comprising: forming a first link together with a second UE in a first frequency band; estimating a second reception timing point related to a second link formed between a third UE and a fourth UE in a second frequency band, based on a synchronization signal and timing advance (TA) information broadcast by the third UE configured to use the second frequency band adjacent to the first frequency band; and determining whether to change a reception timing point of a signal for the first link, based on a difference value between a first reception timing point related to the first link and the second reception timing point.
2. The method according to claim 1, further comprising: transmitting, to the second UE, a signal requesting a change in a timing advance (TA) value for the first link based on a change in the reception timing point of the signal.
3. The method according to claim 1, wherein the reception timing point of the signal is changed when the difference value is greater than or equal to a predetermined threshold.
4. The method according to claim 3, wherein the predetermined threshold is determined based on a cyclic prefix (CP) length of the first signal.
5. The method according to claim 3, wherein the reception timing point of the signal is changed when a timing advance (TA) value included in the TA information is greater than the TA value related to the first link.
6. The method according to claim 1, wherein the reception timing point of the signal is changed, when the difference value is greater than or equal to a predetermined threshold and a reception strength of a signal related to the second link is greater than or equal to a predetermined threshold strength.
7. The method according to claim 6, wherein the reception strength is a received signal strength indicator (RSSI) or a reference signal received power (RSRP).
8. The method according to claim 1, wherein the reception timing point of the signal is changed to a first timing point determined based on the second reception timing point.

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9. The method according to claim 8,
wherein the first timing point is equal to the second
reception timing point.
10. The method according to claim 9,
wherein when a reception strength of a signal related to 5
the second link is less than a predetermined threshold
strength after the reception timing point of the signal
has been changed, the reception timing point of the
signal is changed from the first timing point to a first
reception timing point. 10
11. The method according to claim 8,
wherein when an identifier (ID) change of a user equip-
ment (UE) configured to use the second frequency band
is detected, the reception timing point of the signal is
changed from the first timing point to a first reception 15
timing point.
12. The method according to claim 8,
wherein when a difference value between the first recep-
tion timing point and a reception timing point at which
a signal related to the second link is received is less 20
than a predetermined threshold, the reception timing
point of the signal is changed to the first reception
timing point.
13. A first device comprising:
a radio frequency (RF) transceiver; and 25
a processor connected to the RF transceiver,
wherein the processor is configured to form a first link
together with a second device in a first frequency band,
under control of the RF transceiver, estimate a second

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- reception timing point related to a second link formed
between a third device and a fourth device in a second
frequency band, based on a synchronization signal and
timing advance (TA) information broadcasted by the
third device configured to use the second frequency
band adjacent to the first frequency band, and deter-
mine whether to change a reception timing point of a
signal for the first link, based on a difference value
between a first reception timing point related to the first
link and the second reception timing point.
14. A processor comprising:
a memory configured to store a program therein; and
a processor configured to operate based on the program
stored in the memory,
wherein the processor is configured to form a first link
together with a first user equipment (UE) in a first
frequency band, estimate a second reception timing
point related to a second link formed between a second
UE and a third UE in a second frequency band, based
on a synchronization signal and timing advance (TA)
information broadcasted by the second UE configured
to use the second frequency band adjacent to the first
frequency band, and determine whether to change a
reception timing point of a signal for the first link,
based on a difference value between a first reception
timing point related to the first link and the second
reception timing point.

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